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Chapter 27

GELFAND-P001219

28/$\overline{X}$—85 Gelfand Seminar

W. Müller.

Cheeger / Gromov

X — a complete Riemannian manifold, vol$(X) < \infty$, $|K| < 1$.

[Below K:] sectional curvature.

$$\int p(\Omega) - ?$$

Theorem. X — the interior of a compact manifold with boundary *[Below "interior": (diffeomorphic)]*

$$\Rightarrow 1) \int_X p(\Omega) = \chi(X).$$



Original source details: GELFAND-P001219

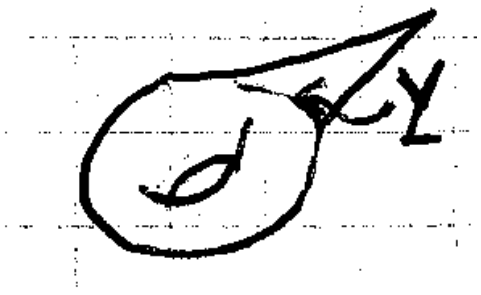
Uncertain reading. An undeciphered mark above $p(\Omega)$ is crossed out.

GELFAND-P001220

$$2) \int_X L_k(p_1, \ldots, p_k) \quad \text{— homotopy invariant}$$

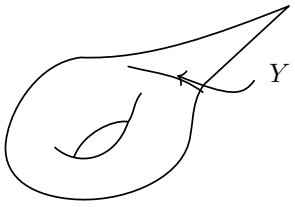
Question related to the work of Cheeger

mfd [inserted above the line X] with cone like singularity



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Diagram and typed description.



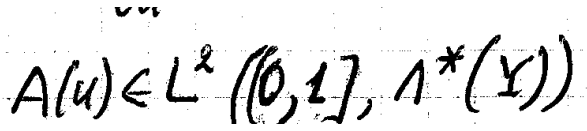
The pointer strokes overlap at the neck.

Δ on Λ^*

Δ on cone

$$-\frac{\partial^2}{\partial u^2} + u^{-2}A(u)$$

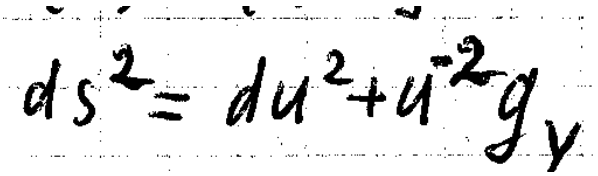
$$A(u) \in L^2((0,1], \Lambda^*(Y))$$



Original source details: GELFAND-P001220

Uncertain reading. Reads $A(u) \in L^2((0,1], \Lambda^*(Y))$. Below, $A(u)$ is written as an operator.

$$A(u) : \Lambda^*(Y) \longrightarrow \Lambda^*(Y),$$
$$C(Y) = (0,1] \times Y,$$
$$ds^2 = du^2 + u^{-2} g_Y.$$



Uncertain reading. The exponent on u before g_Y reads -2.

Original source details: GELFAND-P001220

$$\mathrm{Tr}(e^{-t\Delta}) \underset{t\to 0+}{\sim} \sum_{j\geq 0} a_j t^{-(n-j)/2} + c \log t + b$$

$$b = b_0 \zeta_Y(-1) + \sum_{j\geq 1} b_j \operatorname*{Res}_{s=2j-1} \zeta_Y(s)$$

$$\zeta_Y(s) = \sum_{\lambda_j} \lambda_j^{-s}$$

$$D_s = d + \delta : \Lambda^*_+(X) \longrightarrow \Lambda^*_-(X) \qquad \text{signature operator}$$

[Below:] signature operator

$$(*) \qquad \mathrm{Ind}\, D_s = \int_X L_k(p_1, \ldots, p_k) + \eta_Y(B), \qquad \text{where } B \text{ is defined below.}$$

$\dim X = 4k$.

[Under the bracket at the integral] this would be the index without boundary.

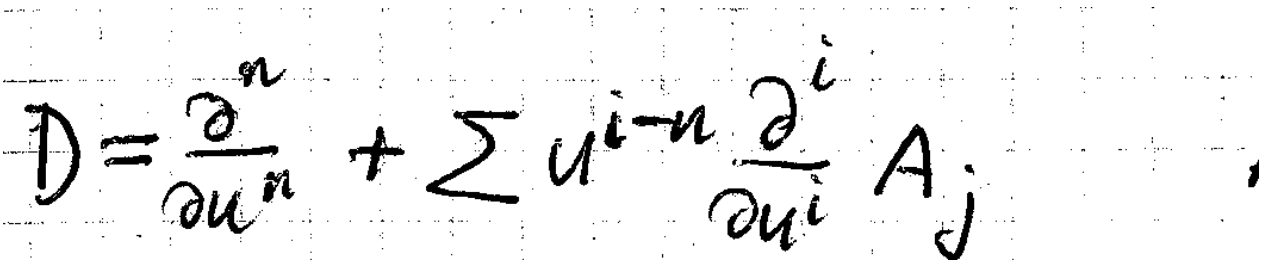
[Enclosed by a bracket] $L_k = 0$ on the cone.

On the cone:

$$d + \delta = D_s = -\frac{\partial}{\partial u} + u^{-1} B, \qquad B : \Lambda^*(Y) \longrightarrow \Lambda^*(Y).$$

[At right, beyond the double vertical line] On Λ^{ev}, B has the form $\pm(*d - d*)$.

$$D = \frac{\partial^n}{\partial u^n} + \sum u^{i-n} \frac{\partial^i}{\partial u^i} A_j$$



Original source details: GELFAND-P001220

Uncertain reading. The powers and derivative in the sum read i. The coefficient reads A_j.

A_j — diff. op. on Y.

index — ?

Cheeger's example gives η! — formula $(*)$

GELFAND-P001221

[source number: 6]

The index is defined on fns. satisfying [crossed out] the condition at the singularity:



Uncertain reading. The crossed-out word before “the condition” is illegible.

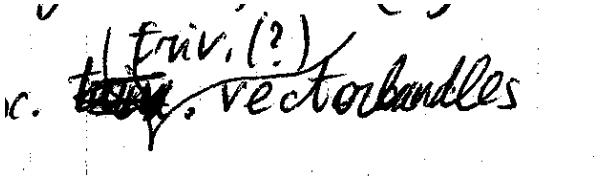
Original source details: GELFAND-P001221

$$\lim_{u\to 0} u^{-1/2}\varphi(u) = \lim_{u\to 0} u^{1/2}\varphi'(u)$$

Relation to representations.

$$X = K\backslash G/\Gamma \qquad \text{loc symm.}, \quad \text{vol}(X) < \infty$$

$\mathcal{E}, \mathcal{F} \longrightarrow X$ loc. [crossed out] [above the line: triv. (?)] vectorbundles



Uncertain reading. “triv. (?)” appears above the crossing-out. The crossed-out letters may also read “triv.”.

Original source details: GELFAND-P001221

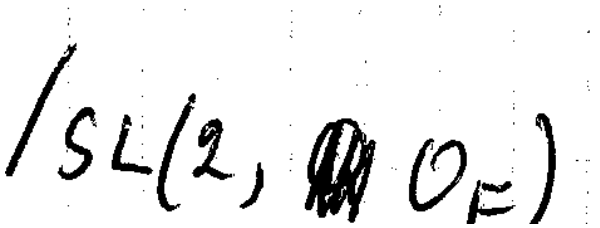
$$D : C_0^\infty(X, \mathcal{E}) \longrightarrow C_0^\infty(X, \mathcal{F}) \qquad \text{oper. loc inv.}$$

Th.

$$\ker \bar{D} < \infty,$$
$$\ker \bar{D}^* < \infty.$$
$$\text{Ind}_{L^2} D = \dim \ker \bar{D} - \dim \ker \bar{D}^*$$

— —

$$X = H \times H/SL(2, O_F), \qquad F = Q(\sqrt{D}), \quad D > 0.$$



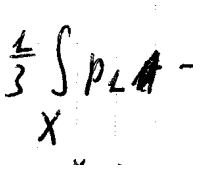
Uncertain reading. A mark before O_F in $SL(2, O_F)$ is heavily crossed out.

Original source details: GELFAND-P001221

only 1 cusp

$$D_s = d + \delta : \Lambda^*_+(X) \longrightarrow \Lambda^*_-(X)$$

$$\operatorname{ind} D_s = \frac{1}{3}\int_X p_1 - \frac{1}{2\pi} L(M, V, 1)\operatorname{vol}(\mathbb{R}^2/M)$$

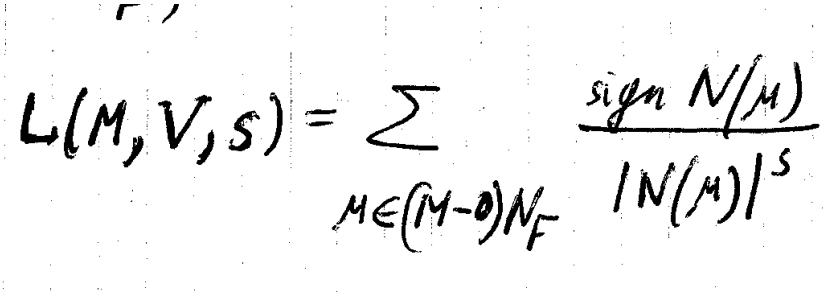


Uncertain reading. The heavily inked mark after p_1 may be an A.

Original source details: GELFAND-P001221

$$M = O_F, \qquad V = O_F^{*2}$$

$$L(M, V, s) = \sum_{\mu \in (M-0)N_F} \frac{\operatorname{sign} N(\mu)}{|N(\mu)|^s}$$



Original source details: GELFAND-P001221

Uncertain reading. The last marks beneath the sum appear to read N_F.

Feigin. On strings (work of Polyakov)

(Continued.)

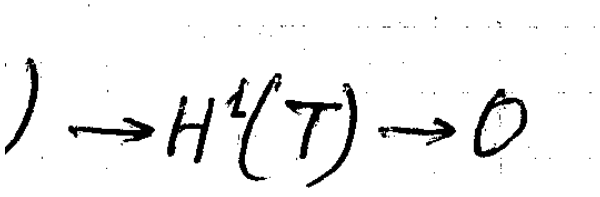
$\mathcal{E}$ — Riem. surface of genus g

T — tangent bundle

On $\mathcal{E}$ a complex structure induced by the Riem. metric is chosen, and the operator $\bar\partial : T^j \longrightarrow T^j \otimes \Omega^{0,1}$;

$$\Delta_j = \bar\partial^* \bar\partial$$

$$0 \longrightarrow H^0(T^j) \longrightarrow \Gamma(T^j) \xrightarrow{\bar\partial} \Gamma(T^j \otimes \Omega^{0,1}) \longrightarrow H^1(T) \longrightarrow 0. \tag{1}$$



Uncertain reading. The term before zero in (1) reads $H^1(T)$, with no exponent on T.

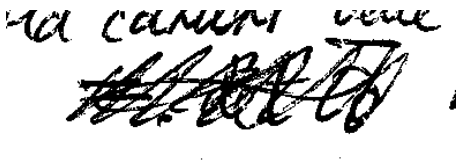
Original source details: GELFAND-P001221

The problem can be reformulated as computing the Ray–Singer torsion of this complex.

$$g_{ij} \longrightarrow g_{ij}(1+\varepsilon\rho) \tag{2}$$

$\bar\partial$ has ker and coker. [In the same interlinear note:] What was actually computed was the change

in Ray–Singer torsion [crossed out] of the complex (1) under deformation



Uncertain reading. The crossed-out notation after "Singer" may read Det $\bar\partial$.

Original source details: GELFAND-P001222

on $\Gamma(T^j)$ and on $\Gamma(T^j \otimes \Omega^{0,1})$ there are Hermitian metrics.

They induce Herm. metrics on $\underbrace{H^0(T^j)}_{\text{subspace}}$ and $H^1(T^j)$.

$\ker \bar\partial\bar\partial^*$ is identified with $H^1(T^j) \Rightarrow$ on $H^1(T^j)$ there is a Herm. metric.

$$\operatorname{Det} T^j := \Lambda^{\max} H^0(T^j) \otimes \left(\Lambda^{\max} H^1(T^j)\right)^* \tag{3}$$

(Quillen's terminol.) (1-dimensional space)

(4) On $\operatorname{Det} T^j$ there is also a Hermitian metric. Denote it by ε.

Ray–Singer torsion is another Herm. metric on $\operatorname{Det} T^j$. It equals the Herm. metric (4) divided by $\operatorname{Det} \Delta_j$:

$$\frac{\varepsilon}{\operatorname{Det} \Delta_j} \tag{5}$$

Consider the variation of the metric:

$$g_{ij} \longrightarrow g_{ij}(1+\varepsilon\rho) \tag{6}$$

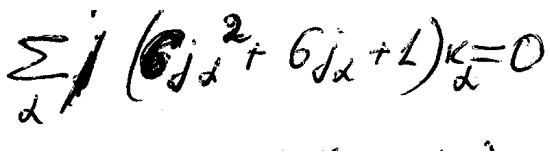
Given a seq. of numbers $j_1, j_2, \dots, j_n$, [inserted above the line: $k_1, \dots, k_n$] and 1-dimensional space

$$\operatorname{Det}(T^{j_1})^{k_1} \otimes \operatorname{Det}(T^{j_2})^{k_2} \otimes \cdots \otimes \operatorname{Det}(T^{j_n})^{k_n} \tag{7}$$

with Herm. metric = the product of the Ray–Singer torsions of each of these spaces.

(8) Claim. The metric on (7) is unchanged under variation (6) if

$$\sum_\alpha (6j_\alpha^2 + 6j_\alpha + 1)k_\alpha = 0. \tag{9}$$



Uncertain reading. There is a heavy diagonal stroke between $\sum_\alpha$ and the opening parenthesis. The first 6 is retraced.

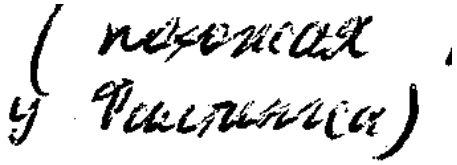
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Or: under condition (9) the metric on (7) depends only on the complex structure.

Let $\mathcal{M}$ — moduli manifold of complex curves of genus g.

The collection of spaces is a 1-dimensional bundle over $\mathcal{M}$.

A metric is constr. on this bundle (a similar constr. occurs in Faltings)



Uncertain reading. The surname at the end of the line is tentatively read as Faltings.

Original source details: GELFAND-P001222

GELFAND-P001223

$4/\overline{\text{XI}} - 85$ Gelfand Sem.

Ryzhikova On lattice representations. (work of Gelfand and Ponomarev)

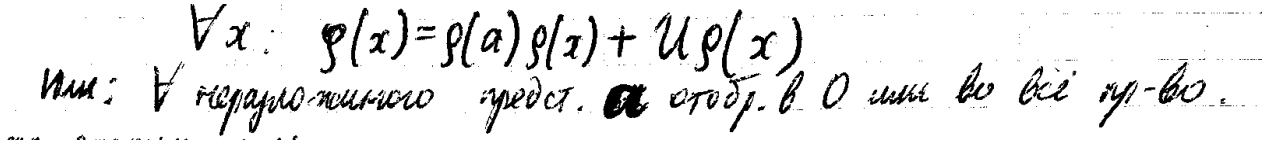
Def. An elem. $a \in L$ is called perfect, if

(L — a lattice)

$$\forall k, \quad \forall \rho : L \longrightarrow \mathcal{L}(V), \quad \exists U \subseteq V,$$
$$U + \rho(a) = V, \qquad U\rho(a) = 0,$$
$$\forall x : \quad \rho(x) = \rho(a)\rho(x) + U\rho(x).$$

[Above k — field. Above ρ — rep..]

Or: $\forall$ indecomposable rep., a maps to 0 or to the whole space.



Original source details: GELFAND-P001223

Uncertain reading. The overwritten sign before "maps" reads a.

Def. of a lattice.

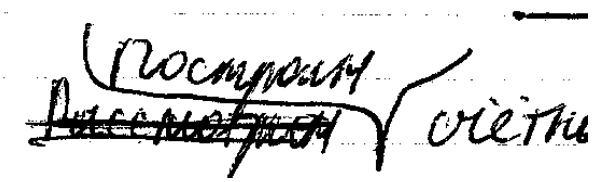
$a + b$, $a \cdot b$ — assoc. commut.

$$a \cdot a = a, \qquad a + a = a, \qquad a(a + b) = a, \qquad a + ab = a,$$
$$a \le b \Longleftrightarrow ab = a \Longleftrightarrow a + b = b,$$
$$a \le b \Longrightarrow b(a + c) = a + bc.$$

Rep. of a lattice $\rho : L \longrightarrow \mathcal{L}(V)$.

— —

[Crossed out: Consider.] [Above the line: Let us construct.] a countable number of cubes $B^+(\ell)$ and $B^-(\ell)$, $\ell \ge 1$,



Uncertain reading. "Let us construct" appears above the crossed-out "Consider".

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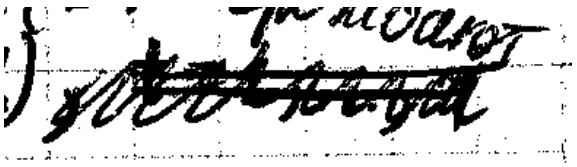
D^r — free modular lattice with r generators $e_1, \ldots, e_r$.

GELFAND-P001224

1. $\ell = 1$

Put $h_t(1) = \sum_{j \neq t} e_j$, $t = 1, \ldots, r$.

These $h_t(1)$ [inserted above the crossing-out: generate] the Boolean algebra $B^+(1)$ [crossed out]



Uncertain reading. The crossed-out text after $B^+(1)$ may begin with "already".

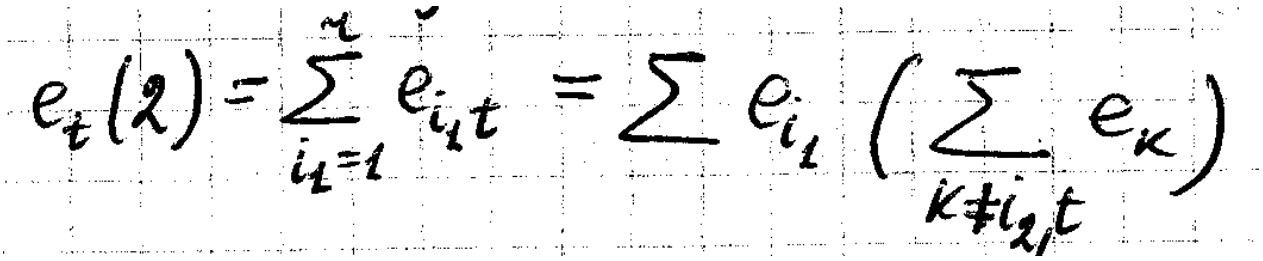
Original source details: GELFAND-P001224

$$0,\ h_1, \ldots, h_4, \quad h_1 \cdot h_2,\ h_2 \cdot h_3,$$
$$h_1 \cdot h_2 \cdot h_3, \ldots,$$
$$h_1 \cdot h_2 \cdot h_3 \cdot h_4.$$

The sum of products is already expr. as a product.

2. $\ell = 2$

$$\{(i_1, i_2) : i_1 \neq i_2\} = A(r, 2)$$
$$\{(i_1, t),\ i_1 \neq t\} = A_t(r, 2) \qquad (t \text{ — fixed})$$
$$e_t(2) = \sum_{i_1=1}^{r} e_{i_1 t} = \sum e_{i_1} \left(\sum_{k \neq i_2, t} e_k \right)$$



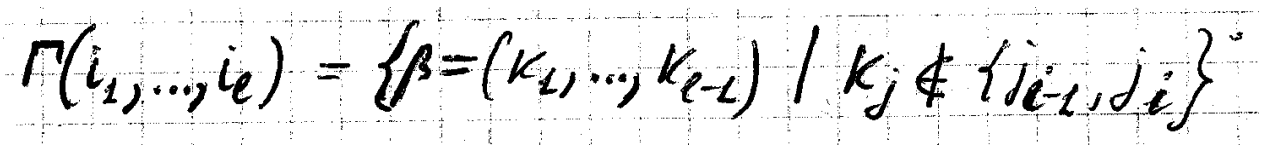
Original source details: GELFAND-P001224

Uncertain reading. The limits of the first sum read $i_1 = 1$ and r. The inner sum reads $k \neq i_2, t$.

$$h_t(2) = \sum_{j \neq t} e_j(2)$$

— —

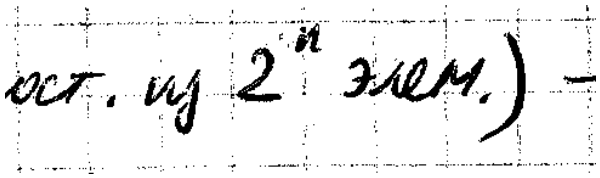
$$\{(i_1, \ldots, i_\ell),\ i_j \neq i_{j+1}\} = A(r, \ell)$$
$$\{(i_1, \ldots, i_{\ell-1}, t),\ i_{\ell-1} \neq t\} \cap A(r, \ell) = A_t(r, \ell)$$
$$\Gamma(i_1, \ldots, i_\ell) = \{\beta = (k_1, \ldots, k_{\ell-1}) \mid k_j \notin \{j_{i-1}, j_i\}$$



Original source details: GELFAND-P001224

Uncertain reading. The final subscripts read $i - 1$ and i. One closing brace is visible.

These are all cubes (consisting of 2^n elem.) — Boolean algebras; the lower ones are constructed dually.

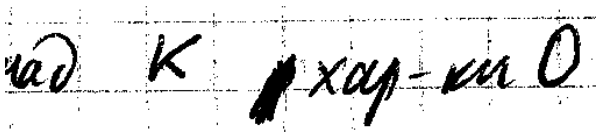


Uncertain reading. The exponent on 2 reads n.

Original source details: GELFAND-P001224

They are all ordered.

Def. V over K [inked over] of char. 0



Uncertain reading. There is a heavily inked mark between K and "of char.".

Original source details: GELFAND-P001224

$\rho : L \longrightarrow \mathcal{L}(V, K)$ is called completely irreducible if

$$\rho(L) \cong \mathcal{L}(\mathbb{Q}^n) \cong P(n-1, \mathbb{Q}).$$

GELFAND-P001225

[source number: 10]

B. L. Feigin On strings

How do the Virasoro alg. and hypergeometric fns. arise?

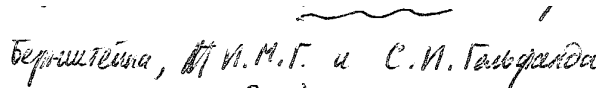
Virasoro algebra: Lie alg. $\mathcal{L}$ with basis $\{e_i\}$, $\underline{\underline{C}}$, $i \in Z$.

$$[e_i, e_j] = (j-i)e_{i+j} + \delta_{i,-j}\frac{i^3 - i}{12}\underline{\underline{C}}, \qquad (1)$$
$$[e_i, \underline{\underline{C}}] = 0.$$

(2) Quotient by the center = alg. of vector fields on the circle.

For Verma modules see the book of Dixmier on enveloping algebras.

and 2 papers by Bernstein, [crossed out] I. M. G. and S. I. Gelfand in FA



Uncertain reading. The crossed-out letter before "I. M. G." may be "И".

Original source details: GELFAND-P001225

(including on categories of Ore)



Uncertain reading. The word after "categories" reads "Ope".

Original source details: GELFAND-P001225

$$\mathfrak{h} = \{e_0, \underline{\underline{C}}\} \qquad \text{(Cartan)},$$
$$\mathcal{L}_+ = \{e_1, e_2, e_3, \ldots\} \qquad \text{(creat.)},$$
$$\mathcal{L}_- = \{e_{-1}, e_{-2}, \ldots\} \qquad \text{(annih.)}.$$

$\chi = (h, \bar{\bar{c}})$ — [crossed out] pair of numbers



Uncertain reading. The crossed-out abbreviation before “pair” may read “ord.”.

Original source details: GELFAND-P001225

Verma module: M_χ, there is a vac. vector v

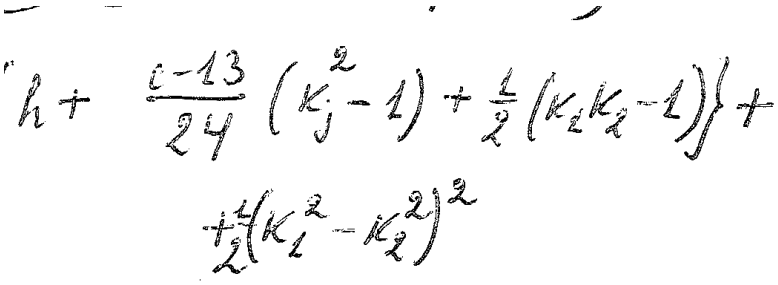
$$\begin{cases} e_0 v = hv, \\ Cv = \bar{\bar{c}}\, v \end{cases} \qquad \mathcal{L}_- v = 0. \tag{3}$$

Physicists write L_i in place of e_i.

(4) Given $M_\chi = M_{h,c}$, when is [inserted above the line: it] reducible?

Physicists call these mass formulas (mass spectrum)

$$0 = \Phi_{k_1 k_2}(h, c) = \prod_{j=1,2} \left\{ h + \frac{c-13}{24}(k_j^2 - 1) + \frac{1}{2}(k_1 k_2 - 1) \right\} + \\ + \frac{1}{2}(k_1^2 - k_2^2)^2. \tag{5}$$



Original source details: GELFAND-P001225

Uncertain reading. The coefficient of $(k_1^2 - k_2^2)^2$ in (5) reads $1/2$.

(Kac formula)

Here k_1, k_2 — natural numbers.

(6) Claim. [Below the heading: (Feigin, Fuchs).] If $\exists\, k_1, k_2$, satisfying (5), the rep. is reducible. If none exist, it is irreducible.

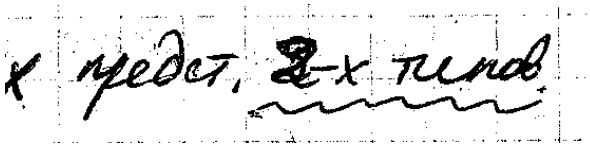
(7) Addendum: if M_χ is reducible, in it $\exists$ a vector different from the vac. vector and annihilated by all annih. operators (a “singular vector”).

$\mathcal{L}$ is naturally graded $\Rightarrow$ M_χ is too.

GELFAND-P001226

(8) Then the level of the sing. vector $= k_1 \cdot k_2$.

Proofs: constructions of various beautiful reps. of 2 types.



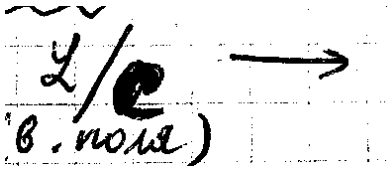
Uncertain reading. The overwritten numeral before “types” reads 2.

Original source details: GELFAND-P001226

1st type.

$$\mathcal{L}/C \longrightarrow gl_\infty,\ O_\infty,\ sp_\infty$$

[Below the quotient: (v. fields).]



Uncertain reading. The overwritten sign in the denominator reads C.

Original source details: GELFAND-P001226

2nd type

(9) $\mathcal{L} \hookrightarrow$ the enveloping alg. of an affine Kac–Moody algebra.

We construct the embedding

$\mathfrak{g}$ — finite-dim. simple Lie alg.

$\widehat{\mathfrak{g}}$ — 1-dimensional extension of the current algebra (usually called a Kac–Moody alg.)

[Between the lines:] ([crossed out] I.M.G. Current alg. — alg. of matrix fns. on the circle).



Uncertain reading. The crossed-out mark before “I.M.G.” is illegible.

Original source details: GELFAND-P001226

On $\mathfrak{g}$ fix the Killing form.

(10) Choose $u_\alpha \in \mathfrak{g}_\alpha$ (root space) so that $(u_\alpha, u_{-\alpha}) = 1$.

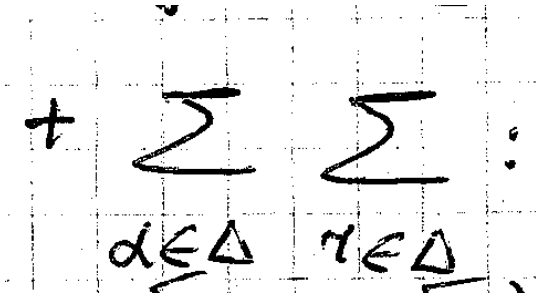
(11) Choose $u_1, \dots, u_\ell \in \mathfrak{h}$, $u^1, \dots, u^\ell$ — dual basis.

[Under $u_1, \dots, u_\ell$: (basis).]

Put

$$S(m) = \sum_{j=1}^{\ell} \sum_{r \in Z} : u^j(-r) u_j(r+m) : + \\ + \sum_{\alpha \in \Delta} \sum_{r \in \Delta} : u_{-\alpha}(-r) u_\alpha(r+m) : \tag{12}$$

(elements of the enveloping alg.)



Original source details: GELFAND-P001226

Uncertain reading. The inner sum on the second line of (12) reads $r \in \Delta$.

The order : : always in $u(r)u(s)$ is set so that $r < s$ or if $r = s$, we replace the product by the half-sum of the products.

$S(m)$ are called Graeme Segal operators.

$$[S(m), S(n)] = 2(n-m) \cdot (g+k) S(m+n) + \\ + \frac{\dim \mathfrak{g}}{12} \delta_{m,-n} (m^3 - m) 4(g+k)k. \tag{13}$$

(k — a number)

$S(m)$ act in a representation of the alg. $\widehat{\mathfrak{g}}$, in which the center acts by multiplication by k (for example, in a Verma module).

GELFAND-P001227

[source number: 11]

g — dual Coxeter number of the algebra $\mathfrak{g}$

(for $n \times n$ matrices this is n)

for $sl(2)$ $g = 2$.

The [crossed out] Virasoro algebra is given by the generators:



Original source details: GELFAND-P001227

Uncertain reading. The crossed-out text before “Virasoro” is illegible. It may start with “Ви...”.

$$e_m = \frac{S(m)}{2(g+k)}. \tag{14}$$

(15) Example. $\mathfrak{g} = sl_2$, $\widehat{\mathfrak{g}} = A_1^1$.

Verma modules are indexed by pairs of numbers (k_1, k_2): notation M_{k_1,k_2}.

The cent. charge in this case is $C = \frac{3k}{k+2}$, where $k = k_1 + k_2$,

$$h = \frac{k_1(k_1+1)}{k+2}.$$

[At right, beyond the vertical line:]

There was a construction: [inserted above "alg.": rep.] of the Virasoro alg. (embedding) is constructed from a rep. of the current alg.

Generalization of the construction of a representation

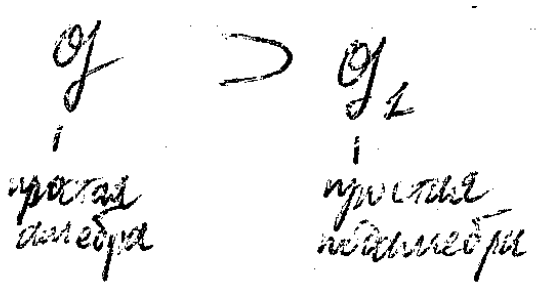


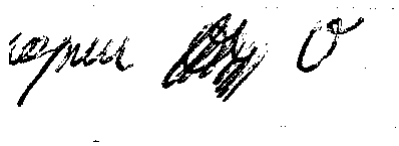
Diagram and typed description.

$\mathfrak{g}$	$\supset$	$\mathfrak{g}_1$
simple algebra		simple subalgebra

Original source details: GELFAND-P001227

$$\widehat{\mathfrak{g}} \supset \widehat{\mathfrak{g}}_1$$

M — a representation of $\widehat{\mathfrak{g}}$ from category [crossed out] O.



Uncertain reading. The mark before O is crossed out and illegible.

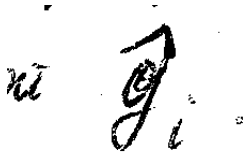
Original source details: GELFAND-P001227

$e_i^{\mathfrak{g}}$, $e_i^{\mathfrak{g}_1}$ — the corresponding e_i, constr. from $\mathfrak{g}$ and $\mathfrak{g}_1$.

Consider $e_i = e_i^{\mathfrak{g}} - e_i^{\mathfrak{g}_1}$.

(16) Theorem. 1) e_i [inserted above the line: also] generate a Virasoro alg.

2) e_i commute with the algebra $\widehat{\mathfrak{g}}_i$.

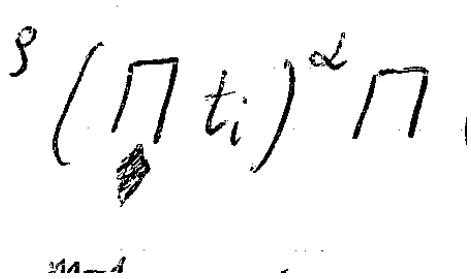


Uncertain reading. The subscript resembles i, with a dot above its main stroke. The same letter takes subscript 1 above.

Original source details: GELFAND-P001227

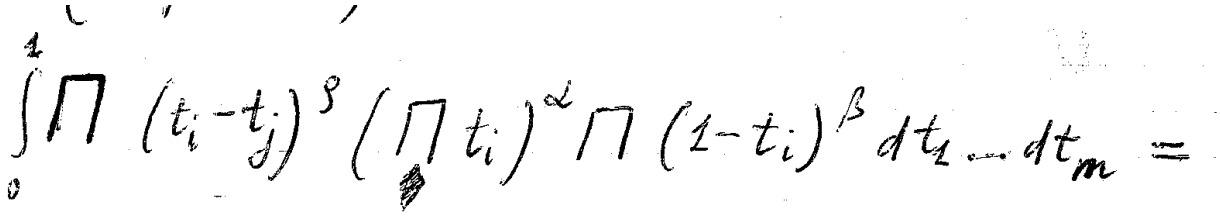
3) Cent. charge of the alg. $\{e_i\}$ = (cent. charge of $e_i^{\mathfrak{g}}$) − (cent. charge of $e_i^{\mathfrak{g}_1}$).
[Under “Cent. charge”, linked by two vertical strokes: (cent. element).]

$$\int_0^1 \prod (t_i - t_j)^\rho \left(\prod t_i\right)^\alpha \prod (1-t_i)^\beta \, dt_1 \dots dt_m = \\ = \prod_{i=1}^{m} \frac{\Gamma(i\rho)}{\Gamma(\rho)} \prod_{i=0}^{m-1} \frac{\Gamma(1+\alpha+i\rho)\,\Gamma(1+\beta+i\rho)}{\Gamma(2+\alpha+\beta+(m-1+i)\rho)}. \tag{17}$$



Uncertain reading. An illegible mark below the second product sign is crossed out.

Original source details: GELFAND-P001227



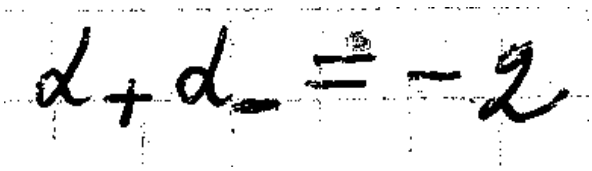
Original source details: GELFAND-P001227

Uncertain reading. The exponent on $(t_i - t_j)$ reads ρ. The first and third product signs have no indices.

Special case of a formula of V. Fateev and Dotsenko.

GELFAND-P001228

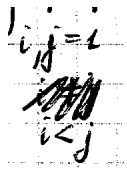
$\alpha_+, \alpha_-, \alpha_+\alpha_- = -2.$



Uncertain reading. The small mark above the equals sign is illegible.

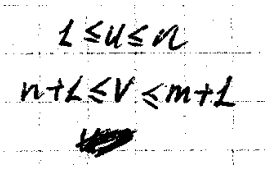
Original source details: GELFAND-P001228

$$\int_0^1 \prod_{\substack{i,j=1\\ i<j}}^{n} (t_i - t_j)^{\alpha_+^2} \prod_{\substack{n+1\le i,j\le n+m\\ i<j}} (t_i - t_j)^{\alpha_-^2} \left(\prod_{i=1}^{n+m} t_i\right)^{\beta} \cdot \prod_{\substack{1\le u\le n\\ n+1\le v\le m+1}} (t_u - t_v)^{-2} \cdot \prod_{1\le i\le n} (z - t_i)^{\gamma} \prod_{n+1\le i\le n+m} (z - t_i)^{\delta}\, dt_1 \dots dt_n. \tag{18}$$



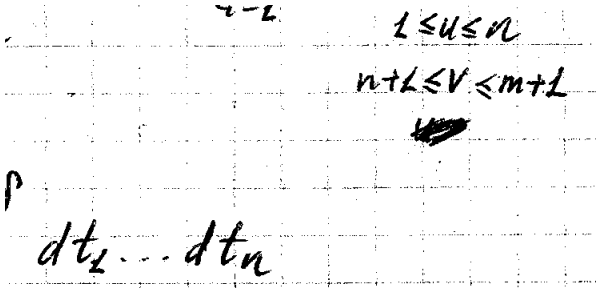
Original source details: GELFAND-P001228

Uncertain reading. Between $i, j = 1$ and $i < j$ beneath the first product, notation involving i and j is crossed out.



Original source details: GELFAND-P001228

Uncertain reading. An illegible mark is crossed out beneath the two constraints on the mixed product.

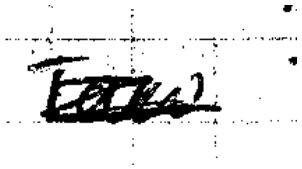


Original source details: GELFAND-P001228

Uncertain reading. The upper bound for v reads $m + 1$, and the final differential reads dt_n.

In this form the 2-point correlation fn. of the theory (conformal field theory) is represented.

[crossed out] This gives a countable number of correlation fns.



Original source details: GELFAND-P001228

Uncertain reading. The crossed-out word before “This gives” may read “If”.



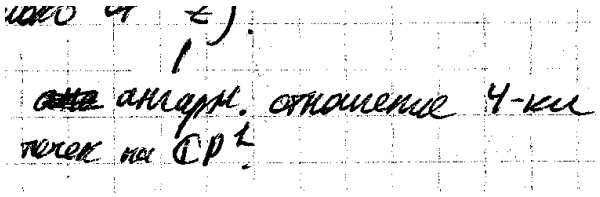
Original source details: GELFAND-P001228

Uncertain reading. The word before “number” is tentatively read as “countable”. The initial strokes are unclear.

The real fns. are their analytic continuation to complex n and m.

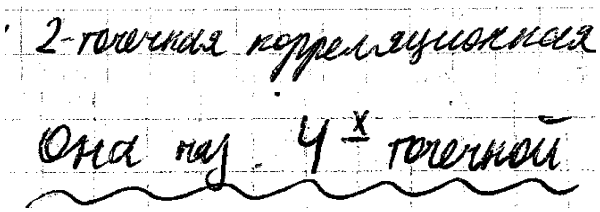
It is called 4-point (although it depends only on z).

[A short stroke leads down from z] [crossed out] cross-ratio of 4 points on CP^1.



Original source details: GELFAND-P001228

Uncertain reading. The crossed-out word before “cross-ratio” may read “it”.



Original source details: GELFAND-P001228

Uncertain reading. The upper passage reads “2-point”. The lower passage reads “4-point”.

GELFAND-P001229

11/XI – 85 Gelfand Sem.

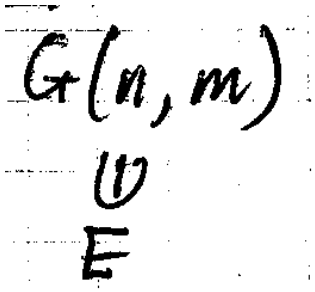
Vershik A. M. Topology of configuration space.



Original source details: GELFAND-P001229

Uncertain reading. The last letter in the word for “space” is overwritten and reads “a”.

1°.



Original source details: GELFAND-P001229

Diagram and typed description.

$G(n, m)$
$\cup$
E

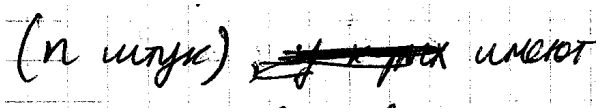
$e_1, \ldots, e_n$ — basis in $\mathbb{R}^n$

$$\tau_E = (Pe_1, \ldots, Pe_n)$$

What is the top. stratum in $G(n, m)$, where τ_E has a given combinat. type?

2°. Combinat. theory of convex polyhedra in $\mathbb{R}^n$.

2 polyhedra V_1, V_2 with ordered (numbered) vertices (n of them) [crossed out] have the same combinat. type if their faces correspond to each other.



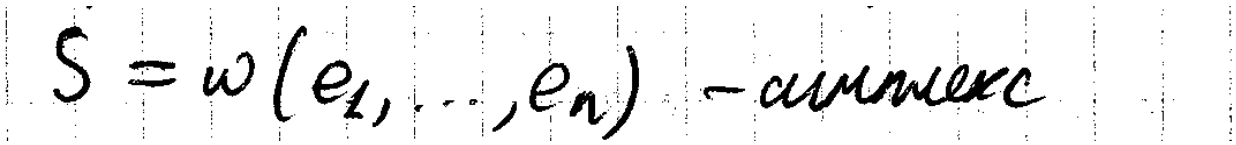
Original source details: GELFAND-P001229

Undeciphered passage. The crossed-out passage after “n of them” is undeciphered.

Grassmannization of the problem.

In $\mathbb{R}^n$ consider a subspace (m-dimensional) E

$$S = \omega(e_1, \ldots, e_n) \quad \text{— simplex}$$



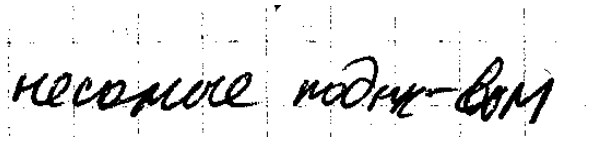
Original source details: GELFAND-P001229

Uncertain reading. The sign after the equality reads ω.

The projection of S onto E — polyhedron (convex)

Ex.: obtain combinatorially any polyhedron of dim. m with n vertices

On $G(n, m)$ consider the partition into subsets where the polyhedra carried by the subspace are combinatorially equivalent.

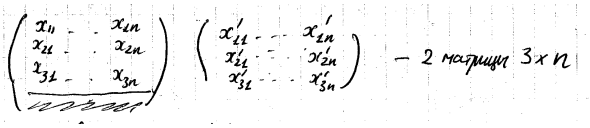


Original source details: GELFAND-P001229

Uncertain reading. The abbreviation after "carried" reads "подпр-вом", for "by the subspace".

What is the topology of these pieces?

3°.



Original source details: GELFAND-P001229

Diagram and typed description.

$$\begin{pmatrix} x_{11} & \cdots & x_{1n} \\ x_{21} & \cdots & x_{2n} \\ x_{31} & \cdots & x_{3n} \end{pmatrix} \qquad \begin{pmatrix} x'_{11} & \cdots & x'_{1n} \\ x'_{21} & \cdots & x'_{2n} \\ x'_{31} & \cdots & x'_{3n} \end{pmatrix}$$

— 2 matrices $3 \times n$

A rule and looping hatching lie under the third row of the first matrix. The second matrix has neither.

Let all minors be $\neq 0$ and have the same signs for x and x'.

Does there exist a homotopy $x(t)$ of matrices with $x(0) = x$, $x(1) = x'$

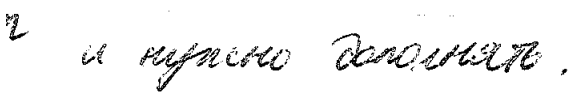
GELFAND-P001230

[source number: 15]

in the same class?

$$n \leq 7 \Rightarrow \text{possible}$$
$$n = 19 \Rightarrow \text{impossible!} \qquad \text{(N. E. Mnëv)}$$

Configurations also make sense in $\mathbb{R}P^n$ and one needs to supplement.



Original source details: GELFAND-P001230

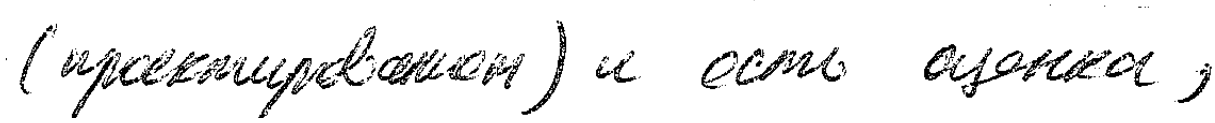
Uncertain reading. The final word reads "дополнять" (to supplement).

One needs to probe the consequences (by projection) and there is a bound on how many one needs to take to have enough.



Uncertain reading. The verb is tentatively read as “пощупать” (to probe).

Original source details: GELFAND-P001230



Original source details: GELFAND-P001230

Uncertain reading. The word in parentheses reads “проектированием” (by projection). The next phrase has “оценка” (bound).

The topology can be that of any algebraic var.!

(This is over $\mathbb{R}$)

over C the analogue is computing π_1.

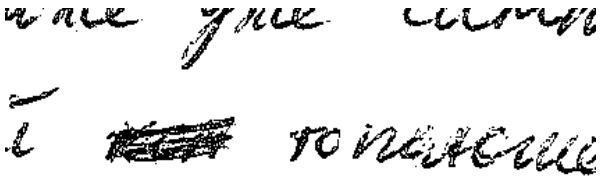
It is interesting what the convex analogue is over C.

Steinitz’s thm.: characterization of the 1-dim. skeleton of a convex polyhedron

(see any textbook of convex geom.) (in $\mathbb{R}^3$)

Corollary: the space of polyhedra of a given combinat. type in $\mathbb{R}^3$ is contractible.

But: in $\mathbb{R}^4$ there already exist even simplicial polyhedra with more or less any [crossed out] topology of combinat. type.



Undeciphered passage. The crossed-out word between “any” and “topology” is undeciphered.

Original source details: GELFAND-P001230

Problem: we want to construct a field of polyhedra on a manifold! (in each tangent sp.). There was [crossed out] a classific. problem for this one.



Undeciphered passage. The crossed-out passage after “There was” is undeciphered.

Original source details: GELFAND-P001230

Matroids m. Given a lattice. It is called a geometric lattice, if it is

1) Semimodular

2) Finite ($\forall$ elem. $=$ sup of finitely many atoms)

[Below this line:] atoms — those immed. after zero.

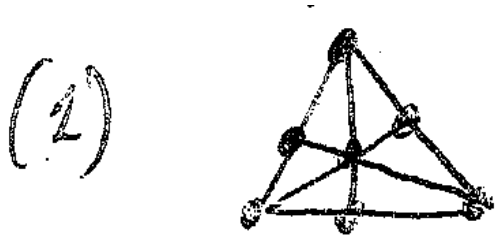


Uncertain reading. The overwritten mark after “Matroids” tentatively reads “м.”.

Original source details: GELFAND-P001230

Config. — geom. lattice.

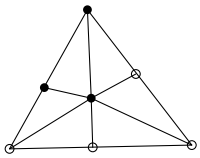
Conversely, let a geom. lattice be given. When is it given by a config.?



Original source details: GELFAND-P001230

Diagram and typed description.

(1)



Seven points and three interior lines are visible. No separate line through the three points on the sides is visible.

(Fano config.) over $\mathbb{R}, \mathbb{Z}$ is not realizable.

A faithful rep. of a lattice = a map from it

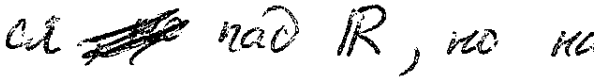
$$a : \Gamma \longrightarrow \mathcal{L}(F^n) \quad \text{— lattice of subspaces,}$$

such that

$$\gamma_1 < \gamma_2 \;\Rightarrow\; a(\gamma_1) < a(\gamma_2)$$

and sup is preserved

(1) is not realizable [crossed out] over $\mathbb{R}$, but over $\mathbb{Z}_2$ realiz..



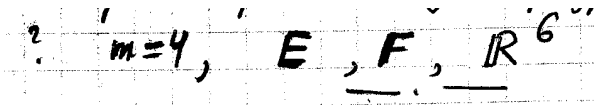
Uncertain reading. The crossed-out mark before “over” is illegible.

Original source details: GELFAND-P001230

GELFAND-P001231

A faithful rep. exists for config. (1):

$$? \qquad m = 4, \quad E, \quad \underline{F}, \quad \underline{\underline{\mathbb{R}^6}}$$



Uncertain reading. The line with the question mark lists $m = 4$, E, F and $\mathbb{R}^6$ separately.

Original source details: GELFAND-P001231

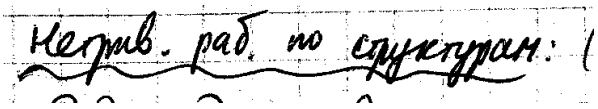
Realizability problem ⇔ problem of solving a system of algebraic inequalities.



Uncertain reading. The final abbreviation reads “нер-в” (inequalities).

Original source details: GELFAND-P001231

Nontriv. work on lattices: (according to Gelfand)



Uncertain reading. The first abbreviation in the heading appears to read “Нетрив.” (nontrivial).

Original source details: GELFAND-P001231

Dedekind: invented it

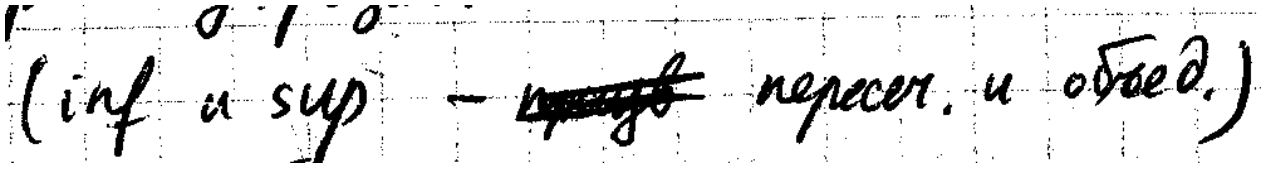
Kurosh, Ore.

von Neumann;

Gelfand, Ponomarev:

[In the left margin: Vershik's add.:] Pudlák and Tůma (Czechs): every finite lattice embeds into the partition lattice of a finite set (very difficult theorem).

(inf and sup — [crossed out] intersection and union)

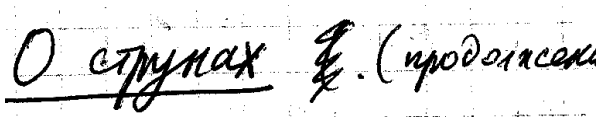


Original source details: GELFAND-P001231

Undeciphered passage. The word after "inf and sup" is crossed out and undeciphered.

0 — partition into points.

Feigin B. On strings [crossed out] (continued).

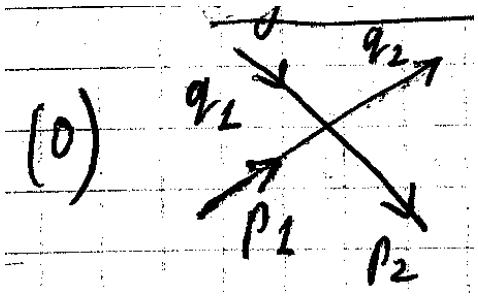


Original source details: GELFAND-P001231

Uncertain reading. The crossed-out vertical mark after "On strings" is undeciphered.

Operator theory of strings.

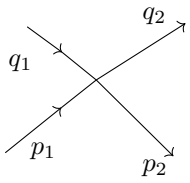
Dual amplitudes. Consider a collision of particles



Original source details: GELFAND-P001231

Diagram and typed description.

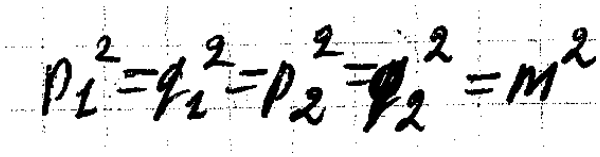
(0)



Let the masses all be the same and the total momentum $= 0$

All are vectors in space/time

$$\begin{cases} p_1 + q_1 = p_2 + q_2, \\ p_1^2 = q_1^2 = p_2^2 = q_2^2 = m^2 \end{cases} \qquad p_j, q_j \in \mathbb{R}^4. \tag{1}$$



Original source details: GELFAND-P001231

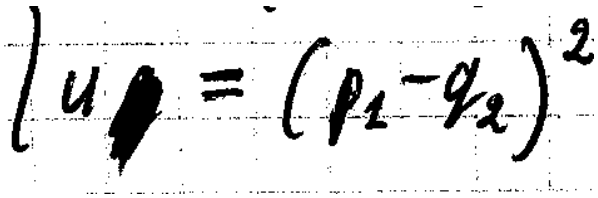
Uncertain reading. The overwritten letter in the penultimate square reads q_2.

Let an S-matrix characterizing this collision be given,

$$S(p_1, q_1, p_2, q_2) \tag{2}$$

Mandelstam variables:

$$\begin{cases} s = (p_1 + q_1)^2, \\ t = (p_1 - p_2)^2 \\ u \text{ [crossed out]} = (p_1 - q_2)^2 \end{cases} \tag{3}$$



Uncertain reading. An illegible mark between u and the equals sign is crossed out.

Original source details: GELFAND-P001231

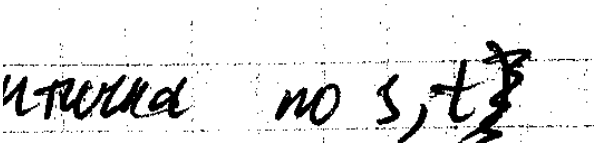
[To the right of (3):] (Relativistically invariant pairwise scalar products are given) dependent:

$$s + t + u = 4m^2 \tag{4}$$

Define $A(s,t)$ (= expression for S in terms of s, t)

Requirement:

$$1) \quad A \text{ is analytic in } s, t \quad \text{[crossed out]} \tag{5}$$



Uncertain reading. A mark after t is crossed out. Its reading is unresolved.

Original source details: GELFAND-P001231

[source number: 16]

2) A is symmetric in s, t (indep. of the order)

3) For fixed t the fn. $A(s,t)$ has in s only simple poles, and [inserted between words: 4)] the locations of the poles do not depend on t. Usually one writes:

$$A(s,t) = \sum_{n=0}^{\infty} \frac{R_n(t)}{n - \alpha(s)} \tag{6}$$

5) $R_n(t)$ — polynomial

The graph of $\alpha(s)$ is called a Regge trajectory.

It turns out that from 1)—5) we uniquely determine the form of the fn. A:

$$A(s,t) = \frac{\Gamma(-\alpha(s))\Gamma(-\alpha(t))}{\Gamma(-\alpha(s) - \alpha(t))} = B(-\alpha(s), -\alpha(t)) \tag{7}$$

It follows from analysis of the exper. data that

$$\alpha(s) \quad \text{— linear fn.} \tag{8}$$

A is called the Veneziano amplitude, or dual amplitude, and conds. 1)—5) are called duality conditions.

Operator method: we want to write $A(s,t)$ as an operator expectation

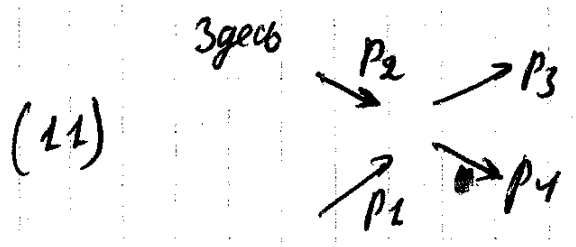
$$\langle 0|V(t)V(s)|0\rangle, \tag{9}$$

where $\langle 0|$ — vac. vector in some space, $V(t)$ — operators.

Another form of (7):

$$S(p_1,p_2,p_3,p_4) = \int_0^1 x^{-2\langle p_1,p_2\rangle}(1-x)^{-2\langle p_2,p_3\rangle}(x(1-x))^{-1-m^2}\,dx \tag{10}$$

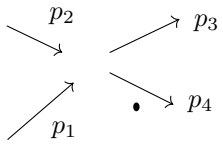
Here



Original source details: GELFAND-P001232

Diagram and typed description.

(11)



A separate ink dot is visible beside the lower-right arrow.

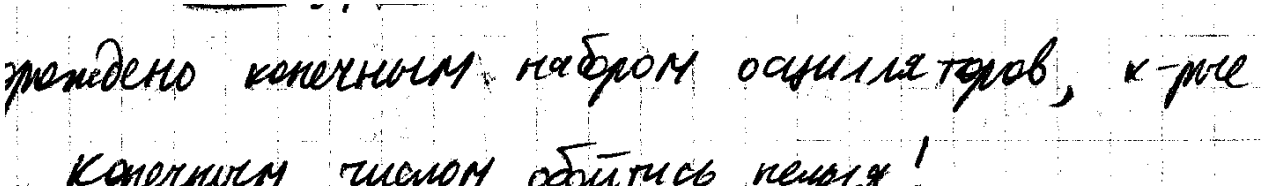
$$p_1 + p_2 + p_3 + p_4 = 0.$$

Similarly one can write amplitudes for a greater number of particles, for example:

$$A(p_1, p_2, p_3, p_4, p_5) \cong \int \int_0^1 x^{\cdots} y^{\cdots} (1-x)^{\cdots} (1-y)^{\cdots} (1-xy)^{\cdots} \, dx \, dy \tag{12}$$

This amplitude has not been calculated in general, only for some special values of the scalar products.

The space is generated by a finite set of oscillators connected as in a string. A finite number will not do!.

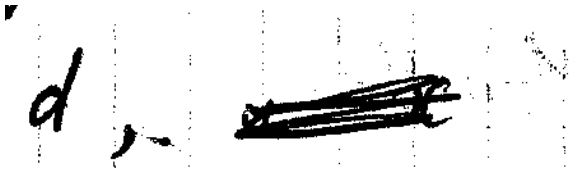


Original source details: GELFAND-P001232

Uncertain reading. The first sentence says “finite”. The second says “A finite number will not do!”

Consider the algebra with generators

$$a_m^\mu, x^\mu, \quad \mu = 0, 1, \ldots, d, \quad \text{[crossed out]} \tag{13}$$



Original source details: GELFAND-P001232

Undeciphered passage. The notation after d in (13) is crossed out and undeciphered.

where d — dim. of the space

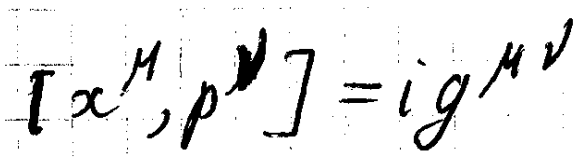
GELFAND-P001233

$$[a_m^\mu, a_n^\nu] = m g^{\mu\nu} \delta_{m,-n} \tag{14}$$

m, n — integers

$$a_0^\mu = p^\mu \tag{15}$$

$$[x^\mu, p^\nu] = i g^{\mu\nu} \tag{16}$$



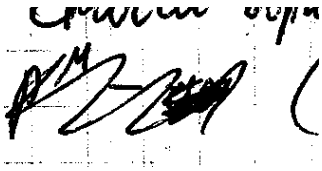
Original source details: GELFAND-P001233

Uncertain reading. The superscript on p in (16) is overwritten and reads ν.

$$[x^\mu, a^\nu_m] = 0, \quad m \neq 0 \tag{17}$$

Meaning of the generators x^μ, a^μ_m:

[Crossed out: p^μ — undeciphered]



Original source details: GELFAND-P001233

Undeciphered passage. The crossed-out continuation after p^μ is undeciphered.

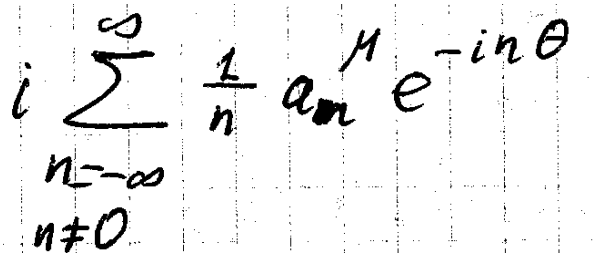
(x^μ, p^μ) — coord. and momentum of the center of gravity of the flying string, a^μ_m — Fourier series expansion of everything else.

Put:

$$P^\mu(\theta) = \sum_{-\infty}^{+\infty} a^\mu_m e^{-im\theta}, \tag{18}$$

(momentum of the point with param. θ)

$$Q^\mu(\theta) = x^\mu + a^\mu_0 \theta + i \sum_{\substack{n=-\infty \\ n\neq 0}}^{\infty} \frac{1}{n} a^\mu_m e^{-in\theta}. \tag{19}$$



Original source details: GELFAND-P001233

Uncertain reading. The subscript on a in (19) is overwritten and tentatively reads m. The summation index is n.

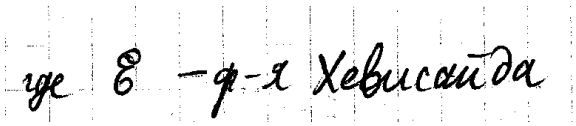
Then the commutation relations hold:

$$[Q_\mu(\theta), P_\nu(\theta')] = 2\pi i g_{\mu\nu} \delta(\theta - \theta') \tag{20}$$

$$[P_\mu(\theta), P_\nu(\theta')] = 2\pi i g_{\mu\nu} \delta'(\theta - \theta') \tag{21}$$

$$[Q_\mu(\theta), Q_\nu(\theta')] = -\pi i g_{\mu\nu} \varepsilon(\theta - \theta'), \tag{22}$$

where ε — Heaviside fn.



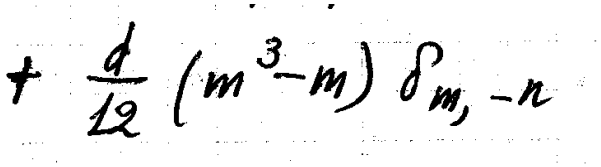
Original source details: GELFAND-P001233

Uncertain reading. The manuscript gives the name “Heaviside” without a definition of ε.

$$L_m \stackrel{\text{def}}{=} \frac{1}{4\pi}\int_0^{2\pi} e^{im\theta} : P^2(\theta) : \, d\theta \tag{23}$$

Then one obtains a representation of the Virasoro algebra

$$[L_m, L_n] = (m-n)L_{m+n} + \frac{d}{12}(m^3 - m)\delta_{m,-n} \tag{24}$$



Uncertain reading. The numerator of the central coefficient reads d.

Original source details: GELFAND-P001233

Exercise: verify this.

Incidentally,

$$L_0 = \frac{1}{2}a_0^2 + \sum_{n=1}^{\infty} a_{-n}a_n \tag{25}$$

GELFAND-P001234

[source number: 17]

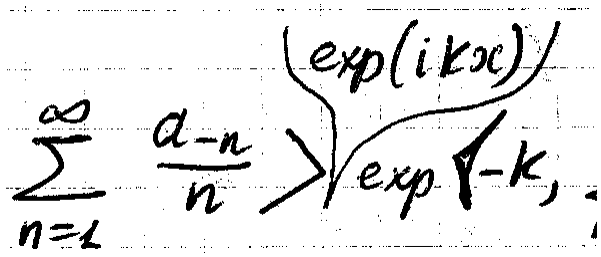
$$2L_m = : \sum_{r+s=m} a_r a_s : \tag{26}$$

Vertex operator: put

$$V(k,\theta) = \exp\langle ik, Q(\theta)\rangle, \tag{27}$$

where k — a vector, then

$$V(k) = V(k,0) = \exp\left\langle k, \sum_{n=1}^{\infty}\frac{a_{-n}}{n}\right\rangle \exp(ikx)\exp\left\langle -k, \sum_{n=1}^{\infty}\frac{a_n}{n}\right\rangle \tag{28}$$



Uncertain reading. The factor $\exp(ikx)$ is inserted above the line between the two exponentials.

Original source details: GELFAND-P001234

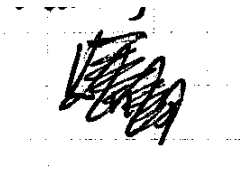
and

$$V(k,\theta) = \exp(i\theta L_0) V(k) \exp(-i\theta L_0). \tag{29}$$

Next,

[Crossed out before (30)]

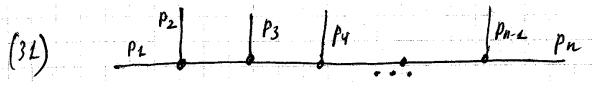
$$[L_n - L_0, V(k)] = \frac{nk^2}{2} V(k). \tag{30}$$



Original source details: GELFAND-P001234

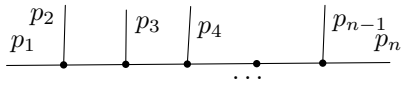
Undeciphered passage. The notation before the commutator in (30) is crossed out and undeciphered.

Consider a series of interactions:



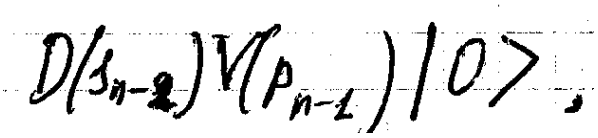
Original source details: GELFAND-P001234

Diagram and typed description.

(31)


and from such a picture construct the amplitude:

$$\langle 0|V(p_2)D(s_2)V(p_3)D(s_3)\dots D(s_{n-2})V(p_{n-1})|0\rangle, \tag{32}$$



Original source details: GELFAND-P001234

Uncertain reading. The overwritten numeral in the last D reads 2, giving the argument s_{n-2}.

where p — the same as k,

$$\begin{aligned} s_2 &= (p_2 + p_1)^2 \\ s_3 &= (p_3 + p_2 + p_1)^2, \\ &\dots \end{aligned} \tag{33}$$

$$D(s) = \int_0^1 x^{L_0 - \alpha(s) - 1} (1-x)^{\alpha(0)-1}\, dx, \tag{34}$$

where $\alpha(s)$ — linear fn. (Regge trajectory).

(32) — dual amplitude. It should agree with A, when there are 4 vertices.

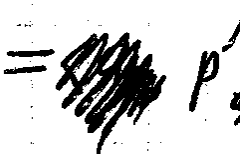
Expand $V(k,\theta)$ in a Fourier series:

$$V(k,\theta) = \sum V_n(k) e^{-in\theta}. \tag{35}$$

There are operators a_i^μ and a_0^μ and vacuum vectors $|0,p\rangle$, satisfying the condition

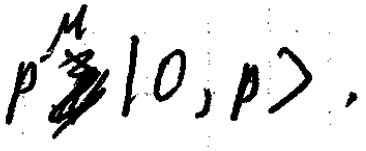
$$a_i^\mu |0,p\rangle = 0, \quad \text{if } i > 0, \tag{36}_i$$

$$a_0^\mu |0,p\rangle = \text{[crossed out] } p^\mu \text{ [crossed out] } |0,p\rangle. \tag{36}_0$$



Uncertain reading. crossed-out writing before p^μ

Original source details: GELFAND-P001235



Uncertain reading. crossed-out sign after p^μ

Original source details: GELFAND-P001235

(the vacua are indexed by momentum vectors)

We obtain the space

$$\bigoplus_p H(p) \tag{37}$$

The vertex operator acts as follows:

$$V_n(k) : H(p) \longrightarrow H(p+k). \tag{38}$$

Commutation relations:

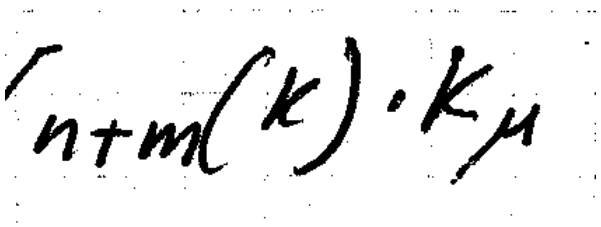
[Crossed out: illegible]



Undeciphered passage. The crossed-out writing under “Commutation relations” is illegible.

Original source details: GELFAND-P001235

$$[a_m^\mu, V_n(k)] = V_{n+m}(k) \cdot k_\mu \tag{39}$$



Uncertain reading. written as k_μ, with a lower index

Original source details: GELFAND-P001235

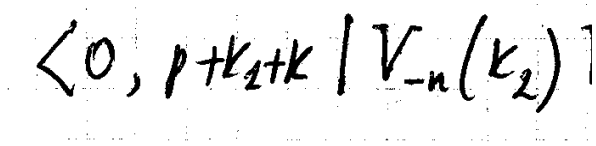
[Crossed out: $V_n(k_1)|0,p\rangle$ / $V_1(k_1)|0,p\rangle$?] Set



Uncertain reading. $V_n(k_1)|0,p\rangle$ / $V_1(k_1)|0,p\rangle$

Original source details: GELFAND-P001235

$$\langle 0, p+k_1+k|V_{-n}(k_2)V_n(k_1)|0,p\rangle = A_n \qquad \text{(these are numbers)} \tag{40}$$



Uncertain reading. k without a subscript

Original source details: GELFAND-P001235

Then

$$\sum A_n z^n = (1-z)^{\langle k_1,k_2\rangle}. \tag{41}$$

This follows by a direct computation. The reader is asked to verify (41) as an exercise.

GELFAND-P001236

[source number: 26]

18 Nov. 1985 Gelfand Seminar

B. Feigin (continued).

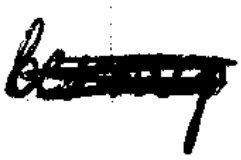
The L_i generate the Virasoro algebra. Let there be a large space on which this algebra acts.

Eigenvectors of L_0 are weight vectors.

By the commutation relations, L_i acts on the space of weight vectors.

(see (24) on p. 17, left)

Highest [crossed out: vector?] weight: a vector that cannot be raised any further



Uncertain reading. vector?

Original source details: GELFAND-P001236

$$\widetilde{\mathcal{H}} = \{v \mid L_0 v = v, \quad L_i v = 0 \text{ for } i > 0\}. \qquad \text{(weight 1)} \tag{1}$$

$$\overline{\mathcal{H}} = \{v \in \widetilde{\mathcal{H}}, \quad v = L_i v', \ i < 0\}, \tag{2}$$

[At right:] Here v' is an arbitrary weight vector from the large space.

(the space of "spurions," a term used by physicists some 15 years ago)

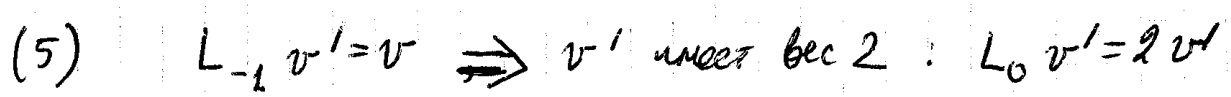
$$\mathcal{H} \stackrel{\text{def}}{=} \widetilde{\mathcal{H}}/\overline{\mathcal{H}} \tag{3}$$

$$\prod_{i=1}^{N} a_{m_i}^{\mu_i} |0, p\rangle \tag{4}$$

[At right:] (space of states)

a large module over the Virasoro algebra

$$L_{-1} v' = v \quad \Rightarrow \quad v' \text{ has weight } 2: \quad L_0 v' = 2v' \tag{5}$$



Original source details: GELFAND-P001236

Uncertain reading. $L_{-1} v' = v; \quad L_0 v' = 2v'$

The aim is to quotient by the diffeomorphism group throughout the space, but a central extension acts, and the centre acts nontrivially.

Another procedure is proposed for obtaining "invariant" vectors. They are invariant under one half, then we quotient by the action of the others and discard them.

The Virasoro algebra disappears, leaving the space of invariants.

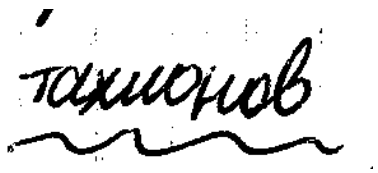
We want to

I. Construct a space generated by $d - 2$ oscillators.

II. Prove that all vectors in $\widetilde{\mathcal{H}}/\overline{\mathcal{H}}$ have norm > 0.

(this does not hold in the original d-dimensional space)

(absence of tachyons[?])



Uncertain reading. tachyons

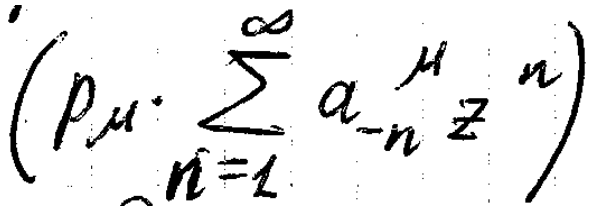
Original source details: GELFAND-P001236

All of this is achieved in dimension 26.

GELFAND-P001237

Vertex operators

$$X(p,z) = \exp\left(p_\mu \cdot \sum_{n=1}^{\infty} a^\mu_{-n} z^n\right) D_p \exp\left(-p_\mu \sum_{n=1}^{\infty} \frac{a^\mu_n}{n} z^{-n}\right), \tag{6}$$



Uncertain reading. $a^\mu_{-n} z^n$ without division by n

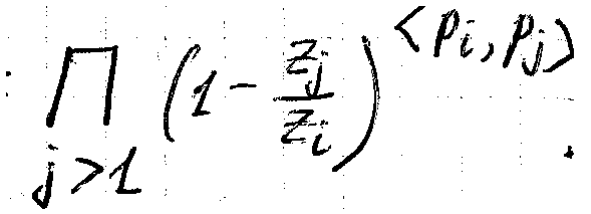
Original source details: GELFAND-P001237

where the operators D_p are defined by the formulas

$$D_p|0,k\rangle = |0,k+p\rangle, \quad [D_p, a^\mu_n] = 0,\ n \neq 0. \tag{7}$$

Correlation function

$$\begin{aligned}&\langle 0, k+\textstyle\sum p_i | X(z_1,p_1)\dots X(z_n,p_n)|0,k\rangle \\ &\quad = \prod_{j>1}\left(1-\frac{z_j}{z_i}\right)^{\langle p_i,p_j\rangle}.\end{aligned} \tag{8}$$



Uncertain reading. the index under the product reads $j > 1$

Original source details: GELFAND-P001237

Expanding $X(z,p)$ in a series gives

$$X(z,p) = \sum X_i(p) z^i \tag{9}$$

(this defines the operators $X_i(p)$)

We want to construct 24 oscillators.

Next we will construct the operators

$$\int X(p,z)\frac{dz}{z}, \tag{10}$$

which commute with the Virasoro algebra when $\langle p,p\rangle = 2$.

$$\begin{aligned} &\text{In } \mathbb{R}^{26} \text{ we want to have } p_1,\dots,p_n: \quad \langle p_i,p_i\rangle = 2 \\ &\langle p_i,p_j\rangle = A_{ij} \in \mathbb{Z}, \quad A_{ij} < 0. \end{aligned} \tag{11}$$

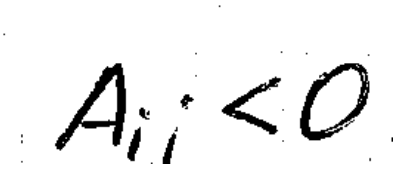
[In the left margin: crossed-out vertical sign?]

Uncertain reading. crossed-out vertical sign



Original source details: GELFAND-P001237

Uncertain reading. $A_{ij} < 0$



Original source details: GELFAND-P001237

Then the operators

$$e_i = \int X(p_i,z)\frac{dz}{z}, \quad f_i = \int X(-p_i,z)\frac{dz}{z} \tag{12}$$

generate an action of the Kac–Moody algebra with Cartan matrix A_{ij} on the space. We must also set

$$h_i|0,k\rangle = \langle k,p_i\rangle|0,k\rangle \tag{13}$$

Read the article by Macdonald (the beginning), Sem. Bourbaki.

Use it to study Lie algebras.

[In the left margin: I.M.G.] The largest finite simple group has $\sim 10^{54}$ elements. It is related to strings!

[Added at right:] (Monster “Moonshine”)

Moonshine

(there is work by Frenkel, Lepowsky and someone else)

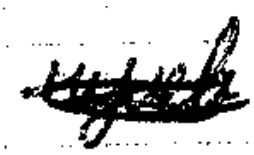
Somehow one obtains a fundamental representation of the Kac–Moody algebra.

GELFAND-P001238

[source number: 27]

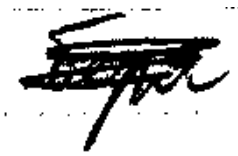
In $\mathbb{R}^{26}$ consider the metric $\langle -1, 1, 1, \ldots, 1\rangle$,

[Crossed out: null?[?]] vector of length 0, [crossed out: illegible]



Uncertain reading. null?

Original source details: GELFAND-P001238



Undeciphered passage. The crossed-out word after “vector of length 0” is illegible.

Original source details: GELFAND-P001238

[Before the next line: two vertical marks, the right one resembling a brace[?]] 24 oscillators



Uncertain reading. two vertical marks, the right one resembling a brace

Original source details: GELFAND-P001238

(Review by D. (?) Schwarz, 1973).



Uncertain reading. D. (?)

Original source details: GELFAND-P001238

— . —

Take $\mathbb{R}^d$ with an indefinite metric and a given vector p with $\langle p, p\rangle = 0$.

[Under “metric”:] (Minkowski, one minus sign)

Set

$$U_n(v) = \int v(z)X(np, z)\frac{dz}{z}, \tag{14}$$

where $v \in \mathbb{R}^d$, $v = \sum v_i e_i$, $v(z) = \sum v_\mu a_n^\mu z^n$

Let

$$v^1, v^2, \ldots, v^{d-2} \text{ be such that } \langle v^i, p\rangle = 0 \tag{15}$$

[“and” inserted over the crossed-out sign:] and [crossed out: crossed-out sign under the inserted “and”[?]]



Uncertain reading. crossed-out sign under the inserted “and”

Original source details: GELFAND-P001238

$v^1, v^2, \ldots, v^{d-2}, p$ are a basis of the space [crossed out: illegible] $\{x : \langle x, p\rangle = 0\}$.



Original source details: GELFAND-P001238

Undeciphered passage. The crossed-out word before the opening brace is illegible.

Set

$$b_n^\mu = U_n(v^\mu); \tag{16}$$

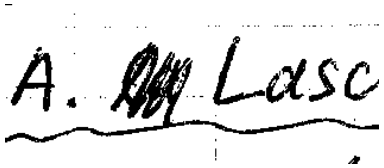
The b_n^μ form a Heisenberg algebra $\Rightarrow$ oscillators.

They generate the whole space.

There is also a review by Green and Schwarz (a volume of Physics Reports).

A. Zelevinsky Schubert cells [inserted above the line: (and polynomials)] and the cohomology of flag varieties.

On the work of A. [crossed out: crossed-out initial?] Lascoux, M.-P. Schützenberger



Original source details: GELFAND-P001238

Uncertain reading. crossed-out initial

(Schubert cells and the cohomology of Grassmannians and flag varieties)

Continuation of work by Bernstein, I.M.G. and S.I.G., Uspekhi Mat. Nauk, 1973, 28(3), 3–26.

X is the set of complete flags in V^n (flags of subspaces).

[Below X, above (1): $\|$, two vertical strokes?]



Original source details: GELFAND-P001238

Uncertain reading. $\|$, two vertical strokes

$$\{V^1 \subset V^2 \subset \ldots \subset V^n\} \tag{1}$$

The cohomology is described in two ways.

1°. Borel

$$H^*(X, \mathbb{Q}) = \mathbb{Q}[x_1, \ldots, x_n]/J$$

The ideal J is generated by symmetric polynomials with zero constant term.

2°. The Schubert cells (c_w) form a basis in homology $H_*(X, \mathbb{Q})$. w is a permutation of the numbers from 1 to n.

$$\begin{matrix} 1 & 2 & \dots & n \\ i_1 & i_2 & \dots & i_n \end{matrix}$$

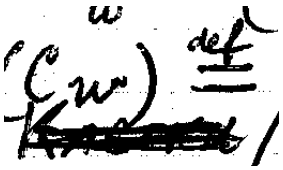
Consider a fixed complete flag

$$F = (\langle e_1\rangle,\ \langle e_1, e_2\rangle,\ \langle e_1, e_2, e_3\rangle, \dots,\ \langle e_1, e_2, \dots, e_n\rangle).$$

Act on it by the permutation w

$$F_w = (\langle e_{w(1)}\rangle,\ \langle e_{w(1)}, e_{w(2)}\rangle, \dots) = (\langle e_{i_1}\rangle,\ \langle e_{i_1}, e_{i_2}\rangle, \dots).$$

[Crossed out: Cell?[?]. Inserted above the crossing-out: $(C_w) \overset{\text{def}}{=}$.]



Uncertain reading. Cell?

Original source details: GELFAND-P001239

[Underlined: (Orbit of the flag F_w under the linear transformations preserving F).] [Crossed out: illegible notation[?].]



Uncertain reading. illegible notation

Original source details: GELFAND-P001239

$$\dim_{\mathbb{C}} c_w = \ell(w) \text{ is the number of inversions in } w.$$

The dual basis in cohomology

$$P_w \in H^{2\ell(w)}(X; \mathbb{Q}).$$

B.–G.–G. found explicit formulas for P_w in the Borel presentation, polynomials (!). [Under the first occurrence of "polynomials": $(\text{mod } J)$.] [Crossed out: illegible[?].]



Uncertain reading. illegible

Original source details: GELFAND-P001239

They have now been lifted to actual polynomials. Their degrees are

$$\deg P_w = \ell(w).$$

$$P_{\mathrm{id}} = 1.$$

$$w_0 = (n, n-1, \dots, 1).$$

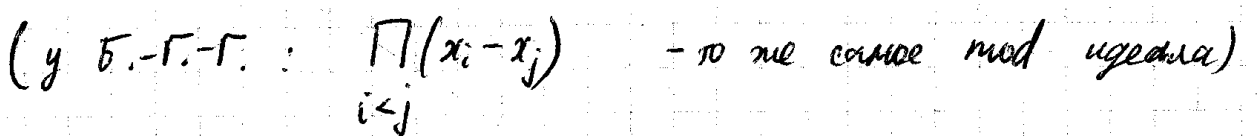
Then

$$\deg P_{w_0} = \frac{n(n-1)}{2}.$$

The choice is nonunique. Take

$$P_{w_0} = x_1^{n-1} x_2^{n-2} \dots x_{n-1}.$$

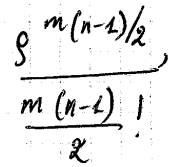
(In B.–G.–G. $\prod_{i<j}(x_i - x_j)$ is the same modulo the ideal.)



Original source details: GELFAND-P001239

Uncertain reading. The product has no normalization factor in the manuscript.

Or $\frac{\rho^{m(n-1)/2}}{\left(\frac{m(n-1)}{2}\right)!}$, where $\rho = (n-1)x_1 + (n-2)x_2 + \dots$



Original source details: GELFAND-P001239

Uncertain reading. The letter before $(n-1)$ reads m both in the exponent and under the fraction bar.

The rest are given by

$$P_w = A_{w_0 w^{-1}} P_{w_0},$$

GELFAND-P001240

[source number: 28]

A_w is the operator corresponding to the permutation

$$w = (i_1, i_1 + 1)(i_2, i_2 + 1) \dots (i_{\ell(w)}, i_{\ell(w)+1})$$

(a product of inversions).

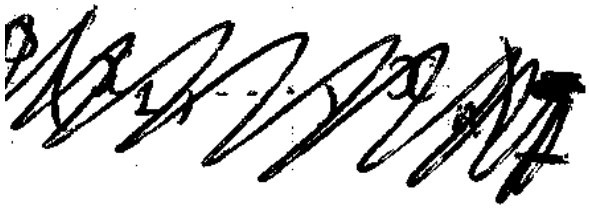


Original source details: GELFAND-P001240

Uncertain reading. In the last factor $+1$ is written at subscript height, $i_{\ell(w)+1}$.

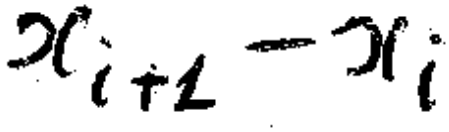
$$A_{(i,i+1)} = \frac{p(x_1, \ldots, x_i, x_{i+1}, \ldots, x_n) - p(x_1, \ldots, x_{i+1}, x_i, \ldots, x_n)}{x_{i+1} - x_i}.$$

[Crossed out after the equals sign, before the fraction: $p(x_1, \ldots$; the rest is illegible.]



Uncertain reading. $p(x_1, \ldots$; the rest is illegible

Original source details: GELFAND-P001240

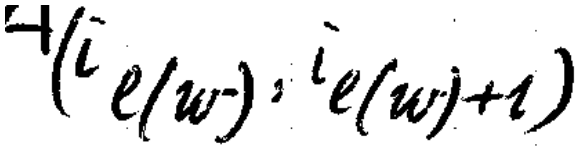


Uncertain reading. The manuscript denominator is $x_{i+1} - x_i$.

Original source details: GELFAND-P001240

The operator

$$A_w = A_{(i_1, i_1+1)} \ldots A_{(i_{\ell(w)}, i_{\ell(w)+1})}$$



Uncertain reading. In the last operator index $+1$ is at the height of $\ell(w)$.

Original source details: GELFAND-P001240

is well defined (independent of the decomposition into a product of inversions). This is the operator used in the formula.

Symmetry condition.

P_w is symmetric in x_i, x_{i+1} if $(i, i+1)$ is not an inversion of w.

GELFAND-P001241

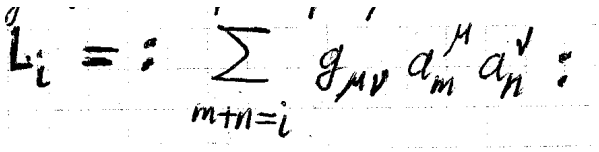
[source number: 35]

25 Nov. 1985 Gelfand Seminar.

Feigin (continued). (On strings).

Generators of the Virasoro algebra

$$L_i =: \sum_{m+n=i} g_{\mu\nu} a_m^\mu a_n^\nu : \tag{1}$$



Uncertain reading. there is no prefactor on the sum

Original source details: GELFAND-P001241

The colons : : make the operator well defined and only add a constant.

I.M.G. What is required for the operation : : to exist? Here it gives rise to the Virasoro algebra. What if we take a lattice or a torus? What happens on the sphere?

$$[L_i, L_j] = (i-j)L_{i+j} + \delta_{i,-j}\frac{i^3 - i}{12}\,c \tag{2}$$

$$p^\mu(t) = \sum a_m^\mu e^{-imt}, \qquad q^\mu(t) = \sum_{m\neq 0} \frac{a_m^\mu}{m} e^{-imt} + q_0^\mu \tag{3}$$

$$\begin{aligned} [x^\mu, a_m^\nu] &= 0, \quad m \neq 0, \\ [x^\mu, a_0^\nu] &= g^{\mu\nu}, \\ x^\mu(t) &= x^\mu + \sum_{n\neq 0} \frac{1}{n}\left(a_n^\mu e^{-int}\right). \end{aligned}$$

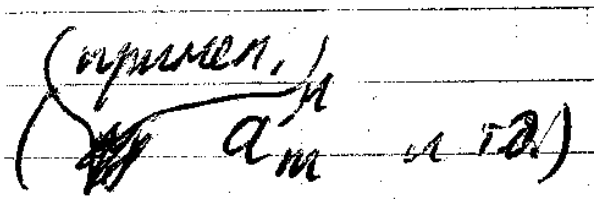
I.M.G. In the classical theory

$$L_n = e^{int}\frac{d}{dt}.$$

Quantum mechanics corresponds to a one-dimensional extension of the classical theory. Compare $[\frac{d}{dx}, x] = 1$.

(4) $\mathcal{H}_p$ is the module generated by the highest vector $|p\rangle$.

[At right: ([crossed out] a_m^μ etc.). "(apply)" is inserted above the crossed-out sign, with a line to the insertion point.]

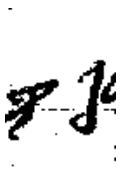


Uncertain reading. illegible crossed-out notation before a_m^μ

Original source details: GELFAND-P001241

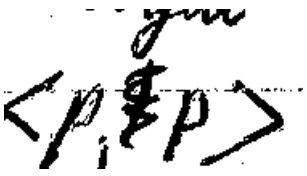
[Between the lines:] This is a module over the Virasoro algebra.

(5) Its structure [crossed out] depends only on $\langle p,$ [illegible] $p\rangle$; by definition



Uncertain reading. small crossed-out sign before "depends"

Original source details: GELFAND-P001241



Uncertain reading. an extra overwritten sign between the comma and the second p

Original source details: GELFAND-P001241

$$a_m^\nu|p\rangle = 0, \quad m > 0, \qquad a_0^\nu|p\rangle = p^\nu|p\rangle \tag{6}$$

(5) $\mathcal{H}_p \cong \mathcal{H}_{p'}$, if $\langle p, p\rangle = \langle p', p'\rangle$. || Denote it by $\mathcal{H}_{\langle p,p\rangle}$.

Consider

$$\bigoplus_{\langle p,p\rangle\in\mathbb{R}} \mathcal{H}_{\langle p,p\rangle} \tag{7}$$

GELFAND-P001242

Define the space of physical states from [crossed out] module (7), as was done [crossed out] above (p. (26)).



Original source details: GELFAND-P001242

Uncertain reading. illegible crossed-out sign before “module”



Original source details: GELFAND-P001242

Uncertain reading. illegible crossed-out sign before “above”

The construction depends [inserted above the line: only] on the [crossed out] action of the Virasoro algebra.



Original source details: GELFAND-P001242

Uncertain reading. crossed-out word, possibly “definition”

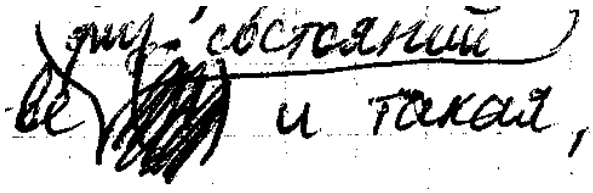
(8) Theorem. (on ghosts).

1) There is a system of oscillators b^{μ}_{m}, $m \neq 0$, $\mu = 1, 2, \ldots, 24$ with standard commutation relations [undeciphered sign] on the space of [crossed out, inserted above the line: physical states], such that



Original source details: GELFAND-P001242

Uncertain reading. sign resembling Cyrillic д



Original source details: GELFAND-P001242

Uncertain reading. illegible crossed-out notation beneath the insertion “physical states”

$$[L_i, b_m^\mu] = 0.$$

The b_m^μ can be constructed from a.

2) $\exists$ a vacuum vector in the space of physical states [inserted above: $\widetilde{\mathcal{H}}$] $|0\rangle$, and $\widetilde{\mathcal{H}}$ is generated by the vectors $|0\rangle$ and the operators b_n^ν. This holds only for $\underline{\underline{d = 26}}$.

3) In $\widetilde{\mathcal{H}}$ there is a natural quadratic form inherited from space (7). It is positive definite on $\widetilde{\mathcal{H}}$.

$$X(p,z) = \exp\left(p_\mu \sum_{n=1}^{\infty} \frac{a_{-n}^\mu}{n} z^n\right) D_p \exp\left(-p_\mu \sum_{n=1}^{\infty} \frac{a_n^\mu}{n} z^{-n}\right), \tag{9}$$

Here D_p is defined by the formula

$$D_p|k\rangle = |k+p\rangle, \tag{10}$$

[Crossed out: and in this case.] Moreover

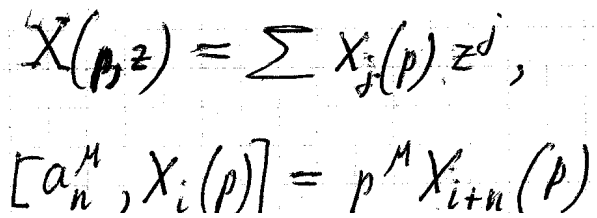
$$[D_p, a_n^\mu] = 0, \quad n \neq 0. \tag{11}$$

Commutation relations for $X(p,z)$ and a_n^μ. If

$$X(p,z) = \sum X_j(p) z^j,$$

then

$$[a_n^\mu, X_i(p)] = p^\mu X_{i+n}(p). \tag{12}$$



Original source details: GELFAND-P001242

Uncertain reading. The expansion reads z^j, and (12) reads $X_{i+n}(p)$.

Representation of the current algebra.

A space of dimension d. In $\mathbb{R}^d$ consider a lattice Q (a discrete subgroup). Assume that Q is integral and unimodular, so there is a basis

$$e_1, \dots, e_d, \quad \text{such that } \langle e_i, e_j \rangle \text{ is integral and } \det \langle e_i, e_j \rangle = 1. \tag{13}$$

Even lattice

$$\forall \gamma \in Q: \quad \langle \gamma, \gamma \rangle \text{ is even.} \tag{14}$$

[Narrow strip of the adjacent leaf at the right edge of the frame.]

Uncertain reading. From top to bottom, four clipped opening parentheses, a short vertical stroke and one more opening parenthesis are visible. Their contents are outside the frame.

Original source details: GELFAND-P001242

GELFAND-P001243

[source number: 36]

(15) Root of a lattice is a vector $\gamma \in Q$, such that $\langle \gamma, \gamma \rangle = 2$.

(16) [Crossed out] Such lattices occur from dimension [crossed out] 8 [crossed out], more precisely, [crossed out] only in dimensions divisible by 8.



Uncertain reading. illegible crossed-out mark after (16)

Original source details: GELFAND-P001243



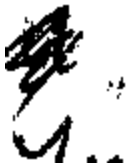
Uncertain reading. illegible crossed-out sign before 8

Original source details: GELFAND-P001243



Uncertain reading. illegible crossed-out sign after 8

Original source details: GELFAND-P001243



Uncertain reading. illegible crossed-out sign before “only”

Original source details: GELFAND-P001243

Exercise Construct such a lattice in $\mathbb{R}^8$.

— . —

Q is a lattice (for now, it need not be even).

Incidentally [Crossed out] The root lattices of groups in series A, D, E have property (14).



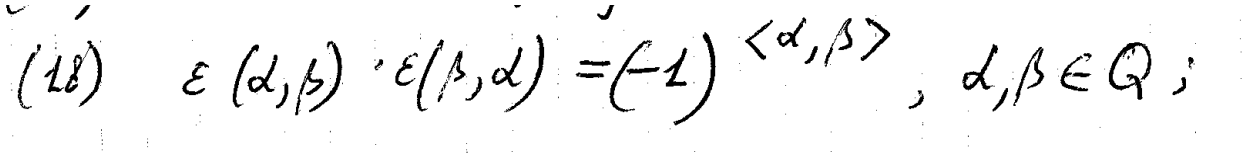
Uncertain reading. crossed-out sign before “root lattices”

Original source details: GELFAND-P001243

Consider the function

$$\varepsilon : Q \times Q \longrightarrow \{\pm 1\} \tag{17}$$

$$\varepsilon(\alpha,\beta)\cdot\varepsilon(\beta,\alpha) = (-1)^{\langle\alpha,\beta\rangle}, \qquad \alpha,\beta \in Q; \tag{18}$$



Original source details: GELFAND-P001243

Uncertain reading. In (18) the condition is written for all $\alpha, \beta \in Q$.

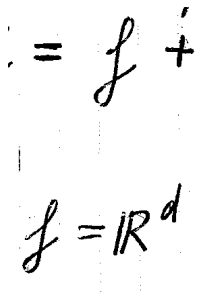
$$\varepsilon(\alpha,\alpha) = -1 \quad \text{if } \langle\alpha,\alpha\rangle = 2. \tag{19}$$

Δ is the set of roots of the lattice Q, i.e. $\Delta = \{\gamma \in Q \mid \langle\gamma,\gamma\rangle = 2\}$.

Let the quadratic form $\langle\ ,\ \rangle$ be positive definite. Introduce the Lie algebra

$$\mathfrak{g} = f + \sum_{\alpha\in\Delta} \mathbb{C}x_\alpha, \tag{20}$$

where $f = \mathbb{R}^d \otimes \mathbb{C} = \mathbb{C}^d$.



Uncertain reading. the two matching looped letters read as f

Original source details: GELFAND-P001243

Define the Lie algebra structure on $\mathfrak{g}$ by the formulas

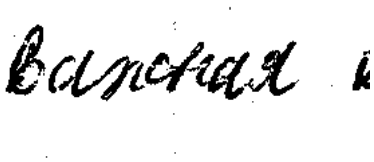
$$[h, x_\alpha] = \langle h,\alpha\rangle x_\alpha; \tag{21}_1$$

$$[x_\alpha, x_\beta] = 0 \quad \text{if } \alpha+\beta \notin \Delta\cup\{0\}; \tag{21}_2$$

$$[x_\alpha, x_\beta] = \varepsilon(\alpha,\beta)x_{\alpha+\beta} \quad \text{if } \alpha+\beta\in\Delta; \tag{21}_3$$

$$[x_\alpha, x_{-\alpha}] = -\alpha. \tag{21}_4$$

This is an important construction of finite-dimensional simple algebras.



Uncertain reading. reads “важная” (important)

Original source details: GELFAND-P001243

Now let the construction above have $p \in Q$, $|p\rangle$ be a vacuum vector, and let $\gamma \in \Delta$. Consider the vertex operator

$$X^\varepsilon(\gamma, z) = X(\gamma, z)\cdot\varepsilon_\gamma; \tag{22}_1$$

where

$$\varepsilon_\gamma|p\rangle = \varepsilon(\gamma,p)|p\rangle, \tag{22}_2$$

$$[\varepsilon_\gamma, a_m^\mu] = 0, \quad m \neq 0. \tag{22}_3$$

Now introduce

$$X^\varepsilon(\gamma, z) = \sum X_i^\varepsilon(\gamma) z^i. \tag{23}$$

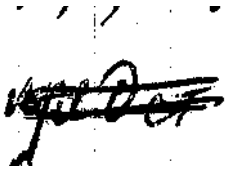
In $\mathbb{R}^{24}$ there is a special (exceptional) lattice that occurs here and in the theory of finite simple groups. There are 26 sporadic simple groups in all.

(24) $\{X_n^\varepsilon(\gamma), a_n^\mu\}$ generate a [inserted above the line: projective] representation of the current algebra on the circle.

$$[X_n^\varepsilon(\gamma_1), X_m^\varepsilon(\gamma_2)] = 0, \quad \text{if } \langle \gamma_1, \gamma_2 \rangle > 0 \tag{25}_1$$

$$[X_n^\varepsilon(\gamma_1), X_m^\varepsilon(\gamma_2)] = \varepsilon(\gamma_1, \gamma_2) X_{n+m}^\varepsilon(\gamma_1 + \gamma_2), \quad \text{if } \langle \gamma_1, \gamma_2 \rangle = -1 \tag{25}_2$$

and other [crossed out] relations can be written down.



Original source details: GELFAND-P001244

Uncertain reading. possibly an abbreviation for “similar”

In $\mathbb{R}^{26}$ choose a vector c, such that $\langle c, c \rangle = 0$, and consider the integrals

$$\int a(z) X(nc, z) \frac{dz}{z} \tag{26}$$

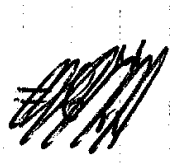
[A superscript on $a(z)$ is crossed out.]



Original source details: GELFAND-P001244

Uncertain reading. possibly μ

[Writing after the integral is crossed out.]



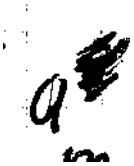
Original source details: GELFAND-P001244

Uncertain reading. the start resembles $=$, and the rest is illegible

Here

$$a(z) = \sum a_m z^m, \tag{27}$$

[A superscript on a_m is crossed out.]



Original source details: GELFAND-P001244

Uncertain reading. possibly μ

where

$$a_m = \sum \alpha_\mu a_m^\mu, \tag{27}'$$

(28) and the vector $(\alpha_1, \dots, \alpha_{26})$ is orthogonal to c.

Set

$$b_n^\mu = \int a^\mu(z) X(nc, z) \frac{dz}{z} \tag{29}$$

$$\langle \alpha, c \rangle = 0. \tag{30}$$

Choose $\alpha^1, \alpha^2, \dots, \alpha^{24}$, so that together with c they form a basis in the space of vectors orthogonal to c. Substituting them in $(27)'$ gives the family a^μ and then b_n^μ.

GELFAND-P001245

[source number: 56]

16 Dec. 1985 Gelfand Seminar

S. P. Novikov.

Soliton theory of the two-dimensional Schrödinger operator.

$n = 2$

$$L = -\partial\bar{\partial} + U = -\Delta + U \tag{1}$$

there are no Lax pairs

$$\frac{\partial L}{\partial t} = [A, L] \quad \Rightarrow \quad \text{dependence on } t \text{ is a one-parameter group of motions of } \mathbb{R}^2 \tag{2}$$

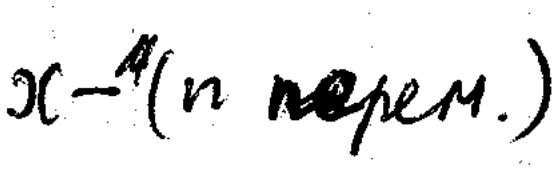
1976 1) Manakov, Uspekhi Mat. Nauk

2) Dubr., Krich., Nov., Dokl. Akad. Nauk.

Difference from the one-dimensional case.

$$U(x) \to 0 \quad \text{as } |x| \to \infty; \quad \Rightarrow \quad \text{there is a scattering amplitude } S(k_1, k_2), \quad |k_1| = |k_2|. \tag{3}$$

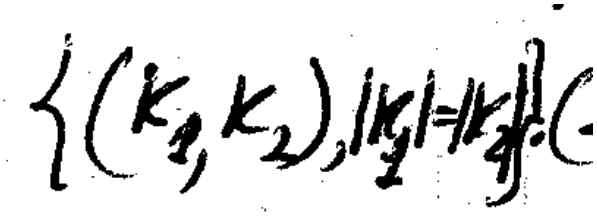
x — [?] (n variables).



Uncertain reading. horizontal stroke after x and an illegible raised sign

Original source details: GELFAND-P001245

[At right:] $\{(k_1, k_2), |k_1| = |k_2|\}$ $(2n-1)$ variables



Uncertain reading. the braces enclose a pair and equality of norms

Original source details: GELFAND-P001245

$$n = 2n - 1 \Leftrightarrow n = 1.$$

An overdetermined problem.

$\underline{n = 2}$ $\Rightarrow$ restriction to fixed $|k_1| = |k_2|$ — also two variables.

Require

$$\begin{cases} \left(\dfrac{\partial L}{\partial t} - [A, L] \right) \psi = 0, \\ L\psi = 0 \end{cases} \qquad (\text{or } L\psi = \varepsilon_0 \psi) \tag{4}$$

or

$$\frac{\partial L}{\partial t} = \underbrace{[A, L] + BL}_{\| \; C_1 L + L C_2}, \qquad A, B \text{ are differential operators.} \tag{5}$$

The inverse problem is solved with the same effectiveness.

Equation (5) is called the two-dimensional analogue of the Lax equation. For $n = 2$ it is now consistent.

$z = x + iy$, $\bar{z}$ and also t.

One should start with periodic problems (or quasiperiodic ones).

Γ is a Riemann surface of genus $g < \infty$, ∞_1, ∞_2 are two points on Γ,

$P_1 + \cdots + P_g$ is a divisor [abbreviation unclear]; w_1, w_2 are local coordinates near ∞_1, ∞_2,

(P_j are other points, $\neq \infty_k$ and distinct from one another), $k_j = w_j^{-1}$.



Uncertain reading. abbreviation before “divisor”, resembling Cyrillic «вхд.»

Original source details: GELFAND-P001245

GELFAND-P001246

(6) Seek an eigenfunction $\Psi(x, y, P)$ with the following properties

[Γ is written below P, with an arrow $\uparrow$ pointing to P.]

1) ψ is meromorphic in P away from ∞_1 and ∞_2. [At right: (x, y and possibly t are parameters).]

2) g poles P_j, independent of (x, y)

3)

$$\psi \sim c_1(x,y)e^{ik_1 z}(1 + O(w_1)), \qquad \text{near } \infty_1$$
$$\psi \sim c_2(x,y)e^{ik_2 \bar{z}}(1 + O(w_2)) \qquad \text{near } \infty_2$$

(7) Gauge transformations

$$\begin{array}{l} c_1 \to cc_1 \\ c_2 \to cc_2 \end{array} \quad \Big| \quad c = c(x, y)$$

change nothing.

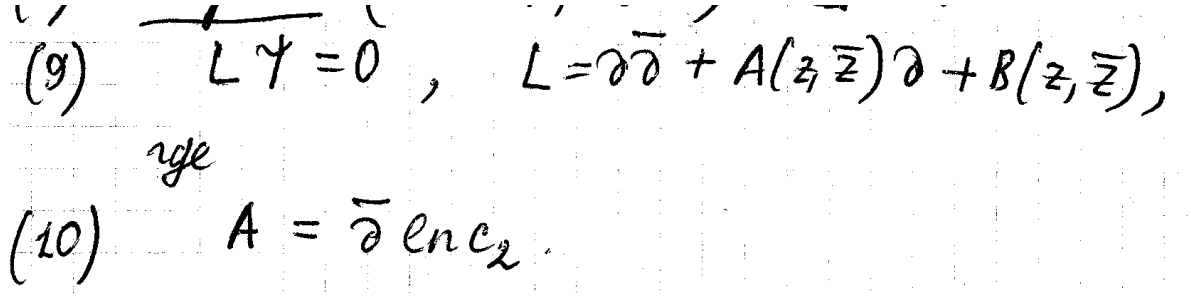
Take $c_1 \equiv 1 \Rightarrow$ it then becomes unique.

(8) Theorem (DKN, 1976) $\exists L$:

$$L\psi = 0, \qquad L = \partial\bar{\partial} + A(z, \bar{z})\partial + B(z, \bar{z}), \tag{9}$$

where

$$A = \bar{\partial} \ln c_2. \tag{10}$$



Original source details: GELFAND-P001246

Uncertain reading. In (9) the derivative after A is ∂ without a bar, and (10) reads $A = \bar{\partial} \ln c_2$.

(11) A, B are quasiperiodic for a Riemann surface in general position.

(12) The magnetic field must have [crossed out] zero flux through the cell (parasitic).



Original source details: GELFAND-P001246

Uncertain reading. crossed-out writing before "zero"

(uninteresting)

(13) Cherednik 1980, Dokl. Akad. Nauk

Veselov, Novikov, Dokl. Akad. Nauk, 1984 (end)

Sufficient conditions [inserted above, with a line to the point after “conditions”: for absence of the first-order term], later found to be necessary. [Below “found”: an arrow pointing left ⟵, a dot and a horizontal stroke.]

The Bloch functions form an n-dimensional complex set.

Consider them at one energy level This is the Fermi surface (complex).

(14) L is called finite-gap [below the line: (or algebro-geometric)] if this surface has finite genus.

P labels the eigenfunction. The P_j are a set of poles (in $\mathbb{C}$, off the spectrum).

∞_j, $j = 1, 2$, are for the operator with constant coefficients.

[Narrow strip of the adjacent leaf at right.]

Uncertain reading. From top to bottom, there is a short vertical stroke, an opening parenthesis, a small clipped mark, another opening parenthesis and a short vertical stroke.

Original source details: GELFAND-P001246

GELFAND-P001247

[source number: 57]

(15) Data for the inverse problem

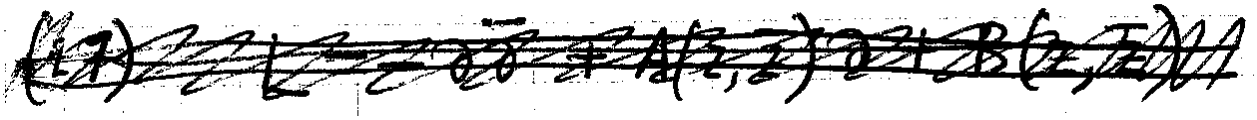
$$\Gamma, \infty_1, \infty_2, w_1, w_2; \qquad D = D_1 + \cdots + D_g$$

construct from them ψ.

(16) Difficulty. There is a first-order term (non-potential).

We want to eliminate the magnetic field. A sufficient condition for a pure potential and reality

[The line numbered (17) is crossed out.]



Original source details: GELFAND-P001247

Uncertain reading. $L = \partial\bar{\partial} + A(z, \bar{z})\partial + B(z, \bar{z})$?

(17) Γ has two involutions

$$\sigma^2 = 1, \quad \tau^2 = 1, \quad \sigma\tau = \tau\sigma \qquad (\text{the group } \mathbb{Z}_2 \times \mathbb{Z}_2)$$

[Below σ: holomorphic. Below τ: antiholomorphic.]

Consider a Riemann surface satisfying (17), and also assume

$$\begin{aligned} \sigma(\infty_i) &= \infty_i, \qquad \sigma(w_i) = -w_i \\ \tau(\infty_1) &= \infty_2, \qquad \tau(w_1) = \pm\bar{w}_2 \end{aligned} \tag{18}$$

$$\tau D = D, \qquad \sigma D + D \cong K + \infty_1 + \infty_2 \tag{19}$$

[A line from $\cong$ points to "linearly equiv."] [A line from K points to "canonical divisor of degree $2g-2$ (divisor of holomorphic differential forms)."]

Hyperelliptic curve.

σ is known (holomorphic). $\psi^*(-k) = \psi(k)$ [A barred sign is written above the equality.]



Original source details: GELFAND-P001247

Uncertain reading. resembles $\bar{\tau}$

(two adjoints, formal and Hermitian)

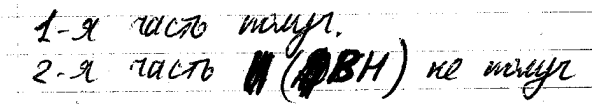
(20) Theorem. Conditions (17)–(19) are necessary and sufficient for reality and for a pure potential.

(21) ψ is expressed using Prym Θ-functions (there are more of them than Jacobi Θ-functions).

Multidimensional case.

Finite-gap property. The family of all Bloch functions (at any energy) contains a nontrivial algebraic curve.

[At right:] First part they write (?). Second part [crossed out] ([illegible]VN) they do not write (?).



Original source details: GELFAND-P001247

Uncertain reading. the word in both lines resembles "пишут" (they write)



Original source details: GELFAND-P001247

Uncertain reading. crossed-out sign after "part"



Original source details: GELFAND-P001247

Uncertain reading. the first letter in parentheses is obscured, the last two read Cyrillic ВН

(22) Example

$$L = \partial\bar{\partial} + V(z,\bar{z}), \qquad V \text{ is a real potential}$$

A direct check gives

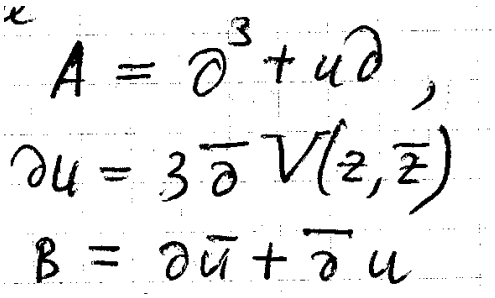
$$\frac{dL}{dt} = [A + \bar{A}, L] + BL,$$

where

$$A = \partial^3 + u\partial,$$

$$\partial u = 3\bar{\partial}V(z,\bar{z}) \qquad \text{(this determines } u\text{)},$$

$$B = \partial\bar{u} + \bar{\partial}u \quad \text{is a scalar function.}$$



Uncertain reading. In the line for u, the first ∂ has no bar and the second has a bar.

Original source details: GELFAND-P001247

[At lower left on the strip of the adjacent leaf.]



Uncertain reading. clipped Cyrillic ending “им)” or “ым)” with a closing parenthesis

Original source details: GELFAND-P001247

GELFAND-P001248

The corresponding system

$$V_t = (\partial^3 + \partial u)V + \text{complex conjugate} \tag{23}$$

there are exact solutions, rational [crossed out] solitons



Uncertain reading. short illegible crossed-out fragment

Original source details: GELFAND-P001248

The rapidly decaying problem is also essentially solved

(R. Novikov, P. Grinevich, Manakov)

D. Knizhnik.

(Dima)

Strings, [crossed out] moduli spaces, quadratic differentials.



Uncertain reading. crossed-out sign before "moduli spaces", possibly "and"

Original source details: GELFAND-P001248

Consider the partition function

$$\begin{aligned} Z_p &= \sum_{\text{surfaces of genus } p} e^{-A} \\ &= \int \mathcal{D}g_{ab}\,\mathcal{D}X_\mu\, e^{-\int \sqrt{g}\, g^{ab}\partial_b X^\mu \partial_a X_\mu \, d^2\xi} \end{aligned} \tag{1}$$

(A is the area)

$\xi^{1,2}$ are coordinates, $g_{ab}(\xi)$ is the metric,

$X^\mu(\xi)$, $\mu = 1, \dots, D$, parametrize the surfaces embedded in $\mathbb{R}^D$.

The action is generally covariant under changes of coordinates, with two arbitrary functions.

(2) Fix some of the degrees of freedom. What remains is $g_{ab} = g(\xi)\delta_{ab}$



Uncertain reading. the factor in (2) reads as $g(\xi)$ with an overwritten letter

Original source details: GELFAND-P001248

$$X^\mu(\xi), \quad \mu = 1, \dots, D = 26$$

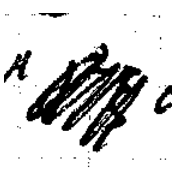
(3) then there is symmetry [inserted above: also] under conformal changes $g_{ab} \to \lambda(\xi) g_{ab}$.

Integrate over the orbits of the gauge transformations.

$D = 26$ $\Rightarrow$ this reduces to integration over the moduli space (Polyakov)

$$\int_{M_p} \frac{\prod dy^i \wedge \overline{dy^i}}{\det(f^i, f^j)} \frac{\int \mathcal{D}X_\mu\, e^{-\int \partial X_\mu \bar\partial X^\mu \, dz\wedge d\bar z}}{\int \mathcal{D}\varepsilon\, e^{-\int \rho\, \partial\bar\varepsilon\, \bar\partial\varepsilon \, dz\wedge d\bar z}} \tag{4}$$

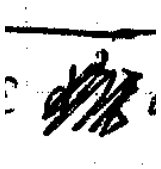
[Writing in the numerator before $dz \wedge d\bar z$ is crossed out.]



Uncertain reading. possibly $d^2\xi$

Original source details: GELFAND-P001248

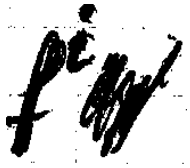
[In the denominator, $d^2\xi$ is crossed out before $dz \wedge d\bar{z}$.]



Uncertain reading. a raised 2 and a sign resembling ξ are visible

Original source details: GELFAND-P001248

f^i [crossed out] are holomorphic [inserted above the line: quadratic] differentials on the curve (a basis)



Uncertain reading. possibly dy^i

Original source details: GELFAND-P001248

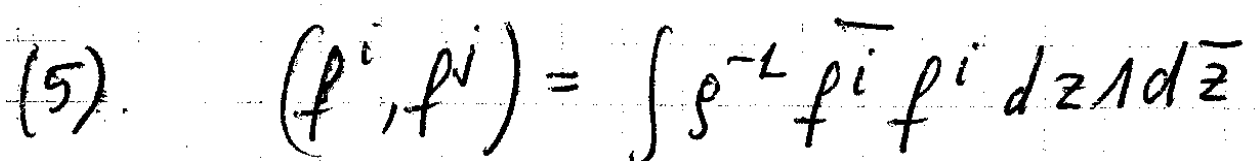
[Writing below "basis" is crossed out.]



Uncertain reading. possibly an abbreviation for "quadratic"

Original source details: GELFAND-P001248

$$(f^i, f^j) = \int \rho^{-1} \overline{f^i} f^i \, dz \wedge d\bar{z} \tag{5}$$



Original source details: GELFAND-P001248

Uncertain reading. Both superscripts on the right of (5) are i, with i, j on the left.

$\rho \, dz \wedge d\bar{z}$ is a fixed metric on the curve (ρ is arbitrary)

$f(dz)^2$ is such a differential

$\varepsilon(dz)^{-1}$ is an invariant quantity.

[At lower right on the strip of the adjacent leaf.]





Uncertain reading. the fragments "(" and, below it, "(1" are visible. The rest is beyond the right edge of the image.

Original source details: GELFAND-P001248

GELFAND-P001249

[source number: 58]

For genus 2 there are three quadratic differentials

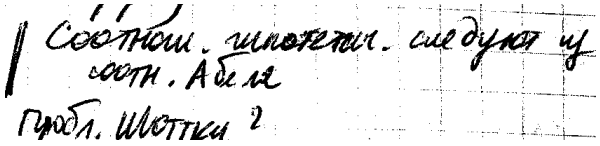
$$\omega_1^2, \quad \omega_2^2, \quad \omega_1\omega_2 \qquad (\omega_1, \omega_2 \text{ are holomorphic differentials})$$

They are always obtained this way, but for genus ≥ 4 there is a surplus.

τ-differentials form a symmetric form.

At most $3p - 3$ quadratic ones (by Riemann–Roch). Choose a dual basis.

[At right, beyond the vertical line:] Relations among products (?) follow from Abel's relations. The Schottky problem?

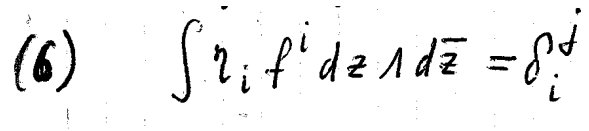


Uncertain reading. the abbreviation resembles Russian «множит.» ("products")

Original source details: GELFAND-P001249

S.P.N. Writing out the relations ⇔ the Schottky problem.

$$\int \eta_i f^i \, dz \wedge d\bar{z} = \delta_i^j \tag{6}$$



Uncertain reading. the index on f is written i, with δ_i^j on the right

Original source details: GELFAND-P001249

[Writing before ρ in (7) is crossed out.]



Undeciphered passage. illegible crossed-out writing before the metric

Original source details: GELFAND-P001249

$$\rho \, dz \, d\bar{z} \longrightarrow \widetilde{\rho} \, |dz + y^i \eta_i \, d\bar{z}|^2 \tag{7}$$

This is a way to parametrize the metric. [Writing after the full stop is crossed out.]



Undeciphered passage. crossed-out writing after "metric"

Original source details: GELFAND-P001249

[At the left edge.]

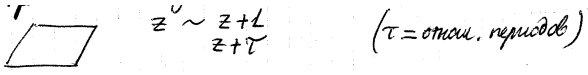


Diagram and typed description. Near the binding is a shape resembling 2, with a vertical zigzag below. Their meaning is unclear.

Original source details: GELFAND-P001249

$p = 1$. The period matrix is a complex number (the ratio of periods along two cycles).

torus ≈ Jacobian



Original source details: GELFAND-P001249

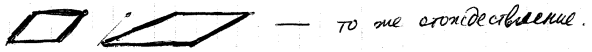
Diagram and typed description.



$z \sim z + 1$
$z + \tau$

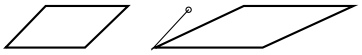
(τ = ratio of periods)

The moduli space is not the whole upper half-plane.



Original source details: GELFAND-P001249

Diagram and typed description.



— the same identification

$$\text{Transformation} \quad \tau \longmapsto \frac{m_1\tau + n_1}{m_2\tau + n_2} \tag{8}$$

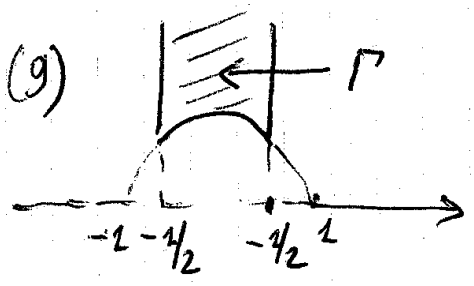
or

$$(\tau_1, \tau_2) \longmapsto (m_1\tau_1 + n_1\tau_2,\ m_2\tau_1 + n_2\tau_2)$$

matrices

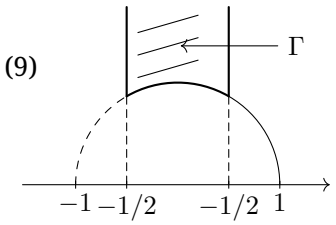
$$\begin{pmatrix} m_1 & n_1 \\ m_2 & n_2 \end{pmatrix}, \quad m_j, n_j \text{ are integers}, \quad m_1n_2 - m_2n_1 = 1$$

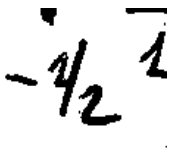
It suffices to use $\tau \longrightarrow \tau + 1$, $\tau \longrightarrow -1/\tau$.



Original source details: GELFAND-P001249

Diagram and typed description.





Original source details: GELFAND-P001249

Uncertain reading. the right-hand vertical is also labelled $-1/2$

$\triangle$ with angles $0°, 60°, 60°$ in the Lobachevsky plane.

Consider integrals over triangle (9).

Choose a basis of holomorphic quadratic differentials on the torus

$$f^1 = 1, \qquad \eta_1 = \frac{1}{\operatorname{Im}\tau} \tag{10}$$

$$du = dz + \delta y^1 \frac{d\bar{z}}{\operatorname{Im}\tau}$$



Uncertain reading. the letter u is heavily overwritten

Original source details: GELFAND-P001249

[Writing before the explanation of u is crossed out.]



Undeciphered passage. illegible crossed-out writing before the new complex symbol

Original source details: GELFAND-P001249

u is a new complex coordinate (complex structure)

y^1 is a new coordinate in moduli space.

The matrix has exactly one entry $\dfrac{dy^1 \wedge d\bar{y}^1}{\operatorname{Im}\tau}$ (11)

GELFAND-P001250

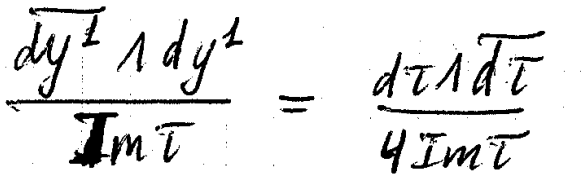
$$\tau \longrightarrow \frac{u(\tau)}{u(1)} = \frac{\tau + \dfrac{\delta y^1 \bar{\tau}}{\operatorname{Im}\tau}}{1 + \dfrac{\delta y^1}{\operatorname{Im}\tau}} \tag{12}$$

$$u(z) = z + \frac{\delta y^1}{\operatorname{Im}\tau}\bar{z} \tag{13}$$

y gives coordinates in the space of complex structures

y^1 is arbitrary, τ is calculated from y^1

$$\frac{\overline{dy^1} \wedge dy^1}{\operatorname{Im}\tau} = \frac{d\tau \wedge d\bar{\tau}}{4\operatorname{Im}\tau}$$



Uncertain reading. the barred differential comes first on the left and second on the right

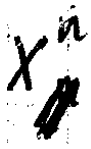
Original source details: GELFAND-P001250

Now it remains to compute

$$\int_\Gamma \frac{d\tau \wedge d\bar{\tau}}{\operatorname{Im}\tau} \frac{\int \mathcal{D}X_\mu \, e^{\int X_\mu \partial\bar{\partial} X^\mu \, dz\wedge d\bar{z}}}{\int \mathcal{D}\varepsilon \, e^{\int \bar{\varepsilon}\partial\bar{\partial}\varepsilon \, dz\wedge d\bar{z}}} \tag{14}$$

[Below the formula, an upward arrow to the measure.] ↑ (This is not the area! of the metric). This is unsurprising, since the starting point was not invariant.

$$X_\mu = a_\mu^n X^n_{\text{[crossed out]}}, \quad \Delta X^n_{\text{[crossed out]}} = \lambda^n X^n_{\text{[crossed out]}} \tag{15}$$



Undeciphered passage. illegible crossed-out subscript on the first X^n

Original source details: GELFAND-P001250



Undeciphered passage. illegible crossed-out subscript on the second X^n

Original source details: GELFAND-P001250



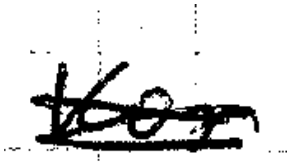
Undeciphered passage. illegible crossed-out subscript on the third X^n

Original source details: GELFAND-P001250

We obtain

$$\frac{\int dX^0_\mu (\det' \Delta)^{-13}}{\int d\varepsilon^0 (\det' \Delta)^{-1}} = \frac{V}{\operatorname{Im}\tau} (\det{}' \Delta)^{-12} \tag{16}$$

[Crossed out: Ko...]



Undeciphered passage. writing after “Ko” is crossed out and illegible

Original source details: GELFAND-P001250

The zero-mode integrals remain.

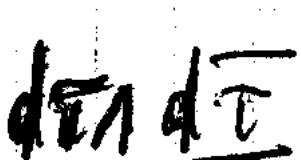
Above is the volume of the universe (26-dimensional).

Below is the area of the torus.

Temporarily set $V = 1$.

Result

$$-\int_\Gamma \frac{d\tau \wedge d\bar{\tau}}{(\operatorname{Im}\tau)^2} (\det{}' \Delta)^{-12} \tag{17}$$



Original source details: GELFAND-P001250

Uncertain reading. the first $d\tau$ is overwritten

[At bottom:] (from m.(?) parametrization) an extra $\operatorname{Im}\tau \Longrightarrow$ we obtain area.



Original source details: GELFAND-P001250

Uncertain reading. the abbreviation resembles Cyrillic «м.»

GELFAND-P001251

[source number: 59]

How to calculate

$$(\det{}' \Delta)^{-1/2} = \int \mathcal{D}\psi \, e^{-\int \partial\psi\bar{\partial}\psi \, dz\wedge d\bar{z}} \tag{18}$$

Calculate the variation of its ln with respect to the variation of complex structure.

Δ is the usual one, for the complex structure!

$$\delta \ln (\det{}' \Delta)^{-1/2} \tag{19}$$

The idea is to use a basis in which it is diagonal, and so on.

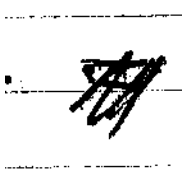
Answer

$$\begin{aligned} (\det{}' \Delta)^{-1/2} &= (\operatorname{Im}\tau)^{-1/2} e^{\frac{\pi}{6}\operatorname{Im}\tau} \prod_{n=1}^{\infty} |1 - e^{2\pi i n \tau}|^{-2} \\ &= (\operatorname{Im}\tau)^{-1/2} |\eta(q)|^{-2} \end{aligned} \tag{20}$$

Use formula (18) for the calculation

$$\delta(\ln \det{}^{-1/2} \Delta) = \frac{\delta \int \mathcal{D}\psi \, e^{-\int \partial\psi\bar{\partial}\psi \, d^2\xi}}{\int \mathcal{D}\psi \, e^{-\int \partial\psi\bar{\partial}\psi \, d^2\xi}} \tag{21}$$

[Writing after the fraction is crossed out.]



Original source details: GELFAND-P001251

Undeciphered passage. illegible crossed-out writing after the fraction in (21)

Here

[Writing before the variation is crossed out.]



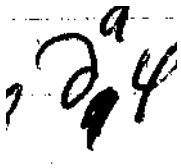
Undeciphered passage. illegible crossed-out writing before the metric variation

Original source details: GELFAND-P001251

$$\delta(\sqrt{g}\, g^{ab}\partial_a\psi\partial_b\psi) = \delta g^{ab} T_{ab},$$

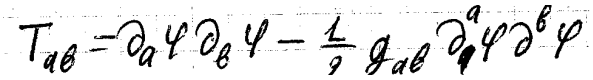
where

$$T_{ab} = \partial_a\psi\partial_b\psi - \frac{1}{2} g_{ab}\partial^a\psi\partial^b\psi$$



Undeciphered passage. illegible crossed-out subscript under the superscript a

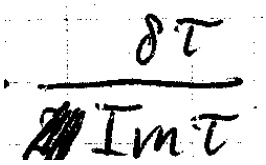
Original source details: GELFAND-P001251



Uncertain reading. the factors after g_{ab} carry superscripts a and b

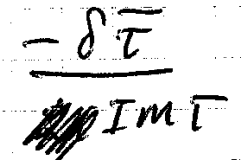
Original source details: GELFAND-P001251

$$\delta g^{++} = \frac{\delta\tau}{\operatorname{Im}\tau}, \qquad \delta g^{--} = \frac{-\delta\bar\tau}{\operatorname{Im}\tau} \qquad (?)$$



Undeciphered passage. crossed-out sign before Im in the denominator of δg^{++}

Original source details: GELFAND-P001251



Undeciphered passage. crossed-out sign before Im in the denominator of δg^{--}

Original source details: GELFAND-P001251

Thus

$$\delta(\ln \overset{-1/2}{\det}\,\Delta) = 2\operatorname{Re}\frac{i}{2}\delta\tau\langle T\rangle \tag{22}$$

where T is the energy–momentum tensor,

$$T(\xi) = \frac{\int \mathcal{D}\psi\, \partial\psi(\xi)\partial\psi(\xi) e^{-\int \partial\psi\bar\partial\psi\, d^2\xi}}{\int \mathcal{D}\psi\, e^{-\int \partial\psi\bar\partial\psi\, d^2\xi}} \tag{23}$$

General formula (on a surface of any genus)

$$\int d^2\xi\, (\eta T + \bar\eta\bar T). \tag{24}$$

[At right, beyond the double vertical line.] $\partial\bar T = \bar\partial T_{+-}$.

(T is an almost holomorphic differential)
For a more precise form of (23), replace $\partial\psi(\xi)\partial\psi(\xi)$ by $:\partial\psi(z)\partial\psi(z):$ and take

$$\begin{aligned} T =: \partial\psi(z)\partial\psi(z) :\overset{\text{def}}{=} \oint \frac{dz'}{2\pi i}\frac{1}{z'-z}\partial\psi(z')\partial\psi(z) \quad \overset{\text{or}}{=} \\ = \langle\partial\psi(z)\partial\psi(z)\rangle \overset{\text{def}}{=} \left\langle \oint \frac{dz'}{2\pi i}\frac{1}{z'-z}\partial\psi(z')\partial\psi(z)\right. \end{aligned} \tag{25}$$

GELFAND-P001252

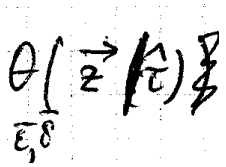
Formula for any genus.
There are p Abelian differentials ω_i and coordinates

$$\vec{z}(Q) = \int_{Q_0}^{Q}\vec{\omega} \tag{26}$$

$$\int_{b_i}\omega_j = \tau_{ij}, \qquad \int_{a_i}\omega_j = \delta_i^j \tag{27}$$

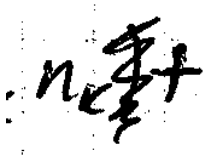
Also introduce θ-functions

$$\theta_{\vec{\varepsilon},\delta}(\vec{z}\mid\hat{\tau}) = \sum_{\vec{n}} e^{\pi i\left(n_i\hat{\tau}_{ik}n_k + 2\pi i\vec{z}\cdot(\vec{n}+\vec{\delta})\right.}$$



Original source details: GELFAND-P001252

Undeciphered passage. illegible crossed-out sign after the arguments of θ



Original source details: GELFAND-P001252

Undeciphered passage. illegible crossed-out sign after $n_i\hat{\tau}_{ik}n_k$ in the first exponent

$$\sum_{\vec{n}} e^{\pi i(\vec{n}+\vec{\delta},\hat{\tau},\vec{n}+\vec{\delta})+2\pi i(\vec{z}+\vec{\varepsilon})(\vec{n}+\vec{\delta})}$$

$$4\vec{\delta}\cdot\vec{\varepsilon} \pmod 2$$

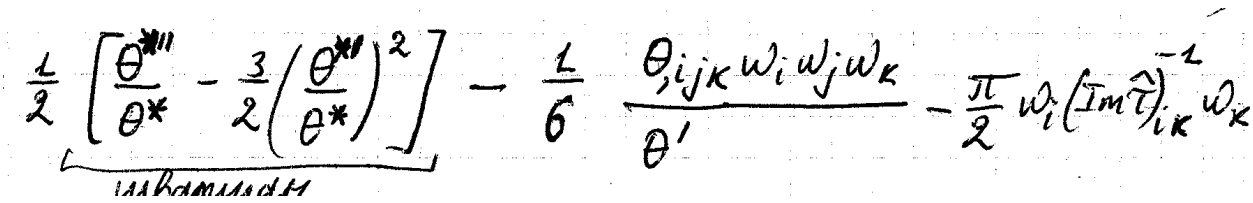
$$\theta^* \equiv \theta^{\bar{\varepsilon},\bar{\delta}}_{,k}\omega_k(z);$$

[Arrow to the coefficient $\theta^{\bar{\varepsilon},\bar{\delta}}_{,k}$.] ⟵ derivative of θ at zero.

[At right:] $\bar{\varepsilon},\bar{\delta}$ is an odd characteristic.

One can take the square root of this Abelian differential (all zeros are of order 2).

$$T = \underbrace{\frac{1}{2}\left[\frac{\theta^{*''}}{\theta^*} - \frac{3}{2}\left(\frac{\theta^{*'}}{\theta^*}\right)^2\right]}_{\text{Schwarzian}} - \frac{1}{6}\frac{\theta_{,ijk}\omega_i\omega_j\omega_k}{\theta'} - \frac{\pi}{2}\omega_i(\operatorname{Im}\hat{\tau})^{-1}_{ik}\omega_k \tag{28}$$



Original source details: GELFAND-P001252

Uncertain reading. the first numerator reads $\theta^{*''}$, and the second denominator reads θ'

All of this together is a quadratic differential.

ω does not include dz. The form is $\omega_i\,dz$.